



Mplify W100 v0.3

LSO SD-WAN Service Schemas and Developer Guide

January 2026

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1 List of Contributing Members

The following members of the Mplify participated in the development of this document and have requested to be included in this list.

Editor Note 1: This list will be finalized before Letter Ballot. Any member that comments in at least one CfC is eligible to be included by opting in before the Letter Ballot is initiated. Note it is the Mplify member that is listed here (typically a company or organization), not their individual representatives.

2 Abstract

This Mplify Standard consists of this schema guide, and its associated software artifacts (JSON/YAML schemas) defines and describes the service-specific payload used in LSO APIs for a set of Service Functions – specifically, Service Order and Service Inventory, for SD-WAN Services. The document starts with an overview of LSO SD-WAN Services. It then provides a basic information model for the Mplify SD-WAN Attributes. The final sections describe the data model focused on the JSON/YAML schemas associated with this specification.

This document can be thought of as a developer's guide for the SD-WAN Services Data Model and the schemas provided that embody the data model. Mplify Services are described by a set of Service Attributes. Each Service Attribute describes an aspect of the service that is agreed between the Service Provider and the Subscriber of the service. The document that describes the Service Attributes for SD-WAN Services is MEF 70.2 [13]. Additional references for the schemas are MEF 105 [15] and MEF 138 [17].

This standard normatively incorporates the following files by reference as if they were part of this document:

https://github.com/MEF-GIT/MEF-LSO/tree/develop_sdWan_service/schema/serviceSchema/sdWan

Schemas for SD-WAN Service Classes:

- schema/serviceSchema/sdWan/sdWanUni.yaml
- schema/serviceSchema/sdWan/swVcEndPoint.yaml
- schema/serviceSchema/sdWan/swVc.yaml
- schema/serviceSchema/sdWan/ucs.yaml
- schema/serviceSchema/sdWan/ucsEndPoint.yaml
- schema/serviceSchema/sdWan/ucsUni.yaml

Schemas for SD-WAN Service Common Classes and Types

- schema/serviceSchema/sdWan/sdWanCommon.yaml
- schema/serviceSchema/sdWan/sdWanTypes.yaml

Schemas for IP Service Common Classes:

- schema/common/common.yaml
- schema/common/ip/ipCommon.yaml
- schema/common/ip/ipRoutingProtocolsCommon.yaml

3 Terminology and Abbreviations

This section defines the terms used in this document. In many cases, the normative definitions of terms are found in other documents. In these cases, the third column is used to provide the reference that is controlling, in other Mplify or external documents. If the reference includes an asterisk (*), the definition has been adapted from the original.

Term	Definition	Reference
Business Applications	The Service Provider functionality supporting Business Management Layer functionality (e.g., product catalog, order management, billing, relationship management, etc.)	MEF 55.1 [11]
Client	The actor in API communication that is making a request.	This document
Information Model	A representation of concepts of interest to an environment in a form that is independent of data repository, data definition language, query language, implementation language, and protocol.	IETF RFC 3444 [5]
Order	One or more Service Order Items formulated into a fulfillment request made by a Client to a Server.	This document
SD-WAN Service Provider	A Service Provider for an SD-WAN Service.	MEF 70.2 [13]
SD-WAN Subscriber	A Subscriber of an SD-WAN Service.	MEF 70.2 [13]
SD-WAN User Network Interface	SD-WAN User Network Interface.	MEF 70.2 [13]
SD-WAN Virtual Connection	An association of SD-WAN Virtual Connection End Points in an SD-WAN Service that provides	MEF 70.2 [13]

Term	Definition	Reference
	the logical construct of a L3 Virtual Private Routed Network for a Subscriber.	
SD-WAN Virtual Connection End Point	A logical construct associated with an SD-WAN UNI where Policies are applied to Ingress and Egress Application Flows.	MEF 70.2 [13]
Server	The actor in API communication which is responding to a request.	This document
Service Attribute	Specific information that is agreed upon between the provider and the user of the service, that describes some aspect of the service behavior or capability.	MEF 70.2 [13]
Service Order	A Buyer/Client submits a Service Order request containing one or more Service Order Items.	MEF 99.1 [14]
Service Inventory	Service Inventory is the inventory managed by the Seller/Server resulting from Service and potentially Product order completion.	MEF 135 [16]
Service Provider	An organization that provides services to Subscribers. In this document, "Service Provider" should read as meaning "SD-WAN Service Provider"	MEF 70.2 [13]
Service Orchestration Functionality	The set of service management layer functionality supporting an agile framework to streamline and automate the service lifecycle in a sustainable fashion for coordinated management supporting design, fulfillment, control, testing, problem management, quality management, usage measurements, security management, analytics, and policy-based	MEF 55.1 [11]

Term	Definition	Reference
	management capabilities providing coordinated end-to-end management and control of services.	
Subscriber	The end-user of a service. In this document, “Subscriber” should be read as meaning “SD-WAN Subscriber”.	MEF 70.2 [13]

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Table 1-Terminology

Term	Definition	Reference
BUS	See Business Applications	MEF 55.1 [11]
SD-WAN	Software Defined Wide Area Network	MEF 70.2 [13]
SD-WAN UNI	SD-WAN User Network Interface	MEF 70.2 [13]
SWVC	SD-WAN Virtual Connection	MEF 70.2 [13]
SWVC EP	SD-WAN Virtual Connection End Point	MEF 70.2 [13]
SOF	Service Orchestration Functionality	MEF 55.1 [11]
UCS	Underlay Connectivity Service	MEF 70.2 [13]
UCS EP	Underlay Connectivity Service End Point	MEF 70.2 [13]
UCS UNI	Underlay Connectivity Service User Network Interface	MEF 70.2 [13]

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Table 2-Abbreviations

4 Compliance Levels

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 (RFC 2119 [2], RFC 8174 [10]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as **[Ox]** for optional.

5 Numerical Prefixes

This document uses the prefix notation to indicate multiplier values as shown in Table 3-Numerical Prefix Conventions.

Decimal		Binary	
Symbol	Value	Symbol	Value
k	10^3	Ki	2^{10}
M	10^6	Mi	2^{20}
G	10^9	Gi	2^{30}
T	10^{12}	Ti	2^{40}
P	10^{15}	Pi	2^{50}
E	10^{18}	Ei	2^{60}
Z	10^{21}	Zi	2^{70}
Y	10^{24}	Yi	2^{80}

Table 3-Numerical Prefix Conventions

6 Introduction

LSO provides a programmatic interface for establishing automated exchange of information (i.e., Service Order, Service Inventory) between a Business Application and Service Orchestration Function. These APIs are hierarchically structured. The outer-most structure includes information relating to the access method (e.g., REST), next is information relating to the function being requested (e.g., Service Order or Service Inventory, etc.), and the inner-most structure contains information relating to the specific service, for example SD-WAN Service).

An SD-WAN Service provides a virtual overlay network that enables application-aware, policy-driven, and orchestrated connectivity between SD-WAN User Network Interfaces (UNIs) and provides the logical construct of a L3 Virtual Private Routed Network for a Subscriber that conveys IP Packets between Subscriber as defined in MEF 70.2 [13].

This specification is accompanied by a data model for SD-WAN Services instantiated as a set of JSON/YAML schemas that can be used within the Service API to perform Service Order, and Service Inventory requests.

The data model for SD-WAN Services includes resource representations for:

- SD-WAN VC: An association of SD-WAN Virtual Connection End Points in an SD-WAN Service that provides the logical construct of a L3 Virtual Private Routed Network for a Subscriber.
- SD-WAN Virtual Connection End Point: A construct that represents the attachment of an SWVC to a UNI. The SWVC End Point provides a container for attributes of the SWVC that can differ at each UNI.
- SD-WAN UNI: A User Network Interface (UNI) is a demarcation point between the responsibility of the SP and the responsibility of the Subscriber. Note that a given UNI always relates to a single SP and a single Subscriber.

The data model for Underlay Connectivity Services includes resource representations for:

- Underlay Connectivity Service: A network service independent of the SD-WAN Service and can have many “characteristics” or “attributes” that define their configuration and behavior.

- Underlay Connectivity Service End Point: A UCS is connected to a UCS UNI by a UCS End Point. Some service architectures support multiple UCSs at the same UCS UNI. The UCS End Point captures Service Attributes at the UCS UNI whose values can differ between UCSs.
- Underlay Connectivity UNI: Access to an Underlay Connectivity Service is provided at the SD-WAN Edge via the UCS UNI.

The document contains the following sections:

- Overview of LSO Architecture (Section 7)
- Overview of SD-WAN Services Model (Section 8)
- Data Model Design Principles and Assumptions (Section 9)
- Data Model for SD-WAN Services (Section 10)
- Relationships between the Entities (Section 11)
- SD-WAN Service Data Model (Section 12)
- Underlay Connectivity Service Data Model (Section 13)
- Common Resources and Types (Section 14)
- IP Common Resources and Types (Section 15)
- IP Common Resources and Types (Section 16)
- SD-WAN Common Resources and Types (Section 17)
- Security Functions for IP Resources and Types (Section 18)
- Usage examples Informative (Appendix A)
- Service Order Operations (Appendix B)

7 Overview of LSO Architecture

MEF 55.1 [11] describes the Reference Architecture for Lifecycle Service Orchestration (LSO) of Mplify-defined connectivity services. MEF 55 defines seven LSO Reference Points that are abstract interconnection points between different domains - either within the Service Provider domain (intra-domain) or between Service Provider and other business entities (inter-domain). One of these LSO Reference Points is LSO Legato that defines the abstract boundary point between a Service Provider's or Partner's Business Application (BA) and Service Orchestration Functionality (SOF) for providing connectivity services provisioning.

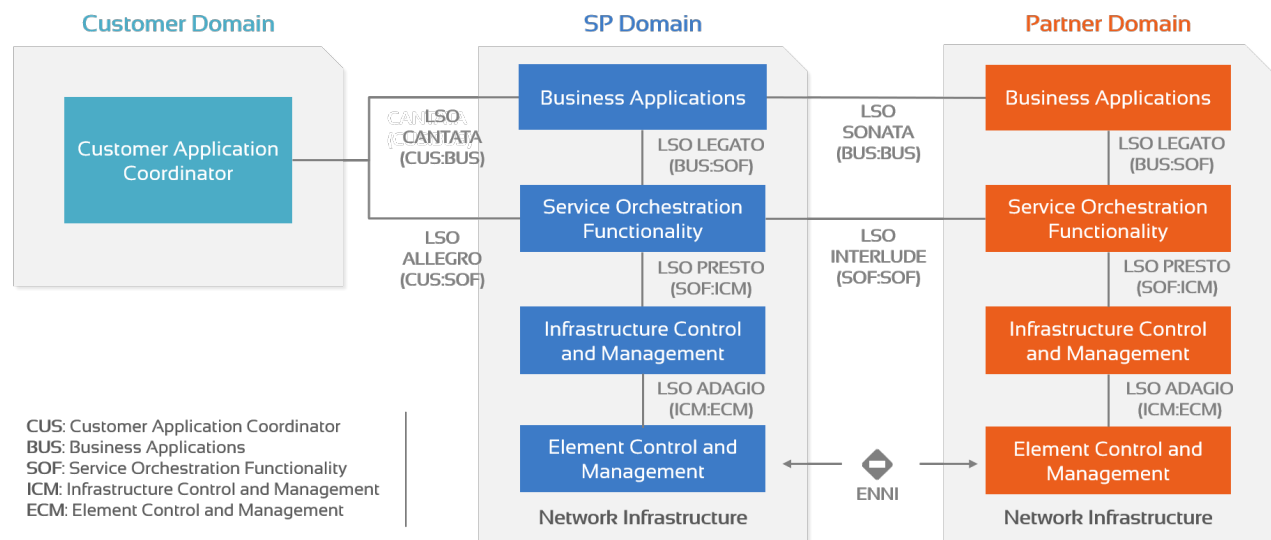


Figure 1-LSO Architecture Diagram

The access to automated service provisioning functionality is provided using the Service Provisioning API at LSO Interface Reference Points (IRPs), Allegro, Interlude and Legato. LSO IRPs provides a suite of APIs for provisioning, inventory, and performance management that are standardized by Mplify as LSO APIs and that are made available by Mplify in a series of releases of the LSO SDK.

The LSO APIs comprise two parts: one is the service-independent functionality, or Basic API Structure also called the Envelope, and the second is the service-specific payload, or Information Payload, as shown in Figure 2.

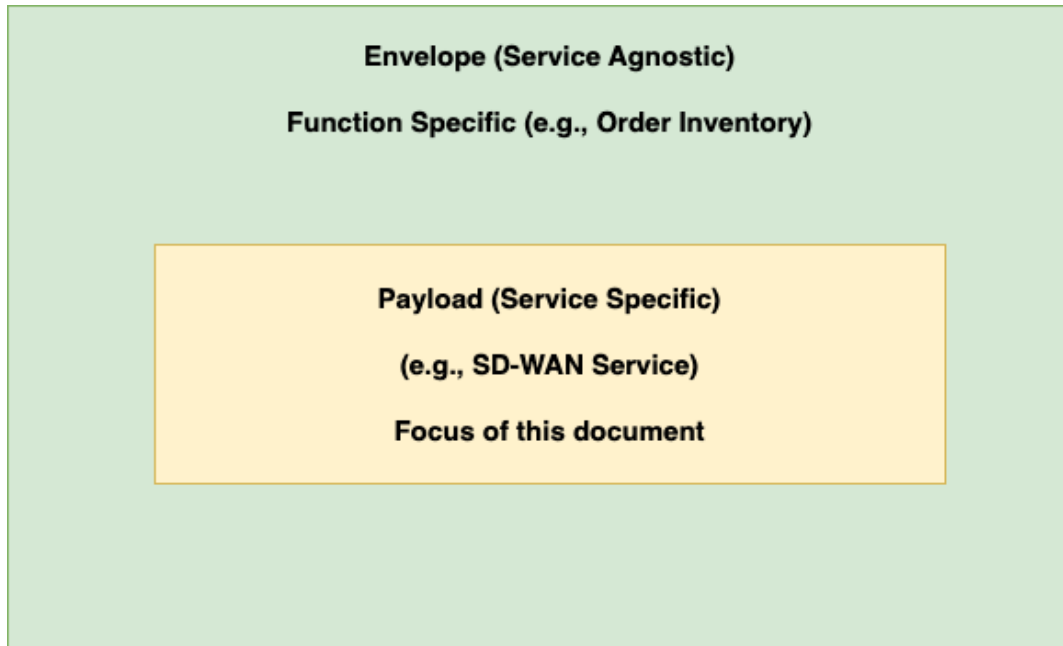


Figure 2-LSO API Envelope and Payload Structure

The main concept behind the API framework is to decouple the common structure, information and operations from the specific service information content. The Envelope part of the API contains the protocol, encoding, security notification mechanisms as well as function-specific information and operations. The Payload part of the API contains the service-specific Payload.

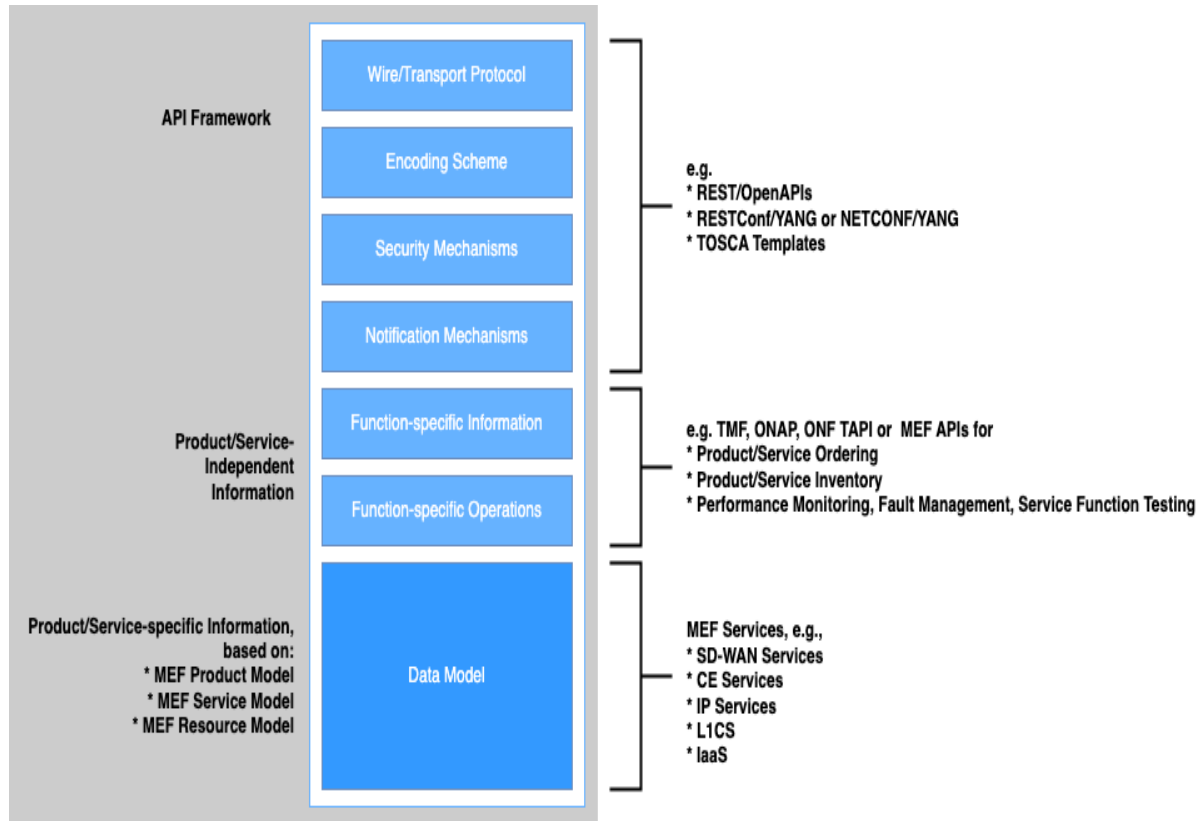


Figure 3-LSO API Structure

This document defines the service-specific payload, shown as JSON data model in Figure 2, for SD-WAN Services as defined in MEF 70.2 [13]. The envelope resources of the API and association to specific payload resources are discussed in detail later in this document.

8 Overview SD-WAN Services Model

The SD-WAN Services model has three main datatypes, SwVc, SdWanUni, and SwVcEndPoint. An SD-WAN Service is an overlay service that operates over one or more Underlay Connectivity Services (UCSs). An SD-WAN Service is defined as having an SwVc and two or more SwVcEndPoints.

A SdWanUni is the demarcation point between the responsibility of the Service Provider and the Subscriber. A UCS has three main classes, Ucs, UcsUni and UcsEndPoint. Figure 4 shows the overview SD-WAN Services Model including the main datatypes for SD-WAN and UCS. Further details for SD-WAN and UCS are provided and included reference to appropriate section(s).

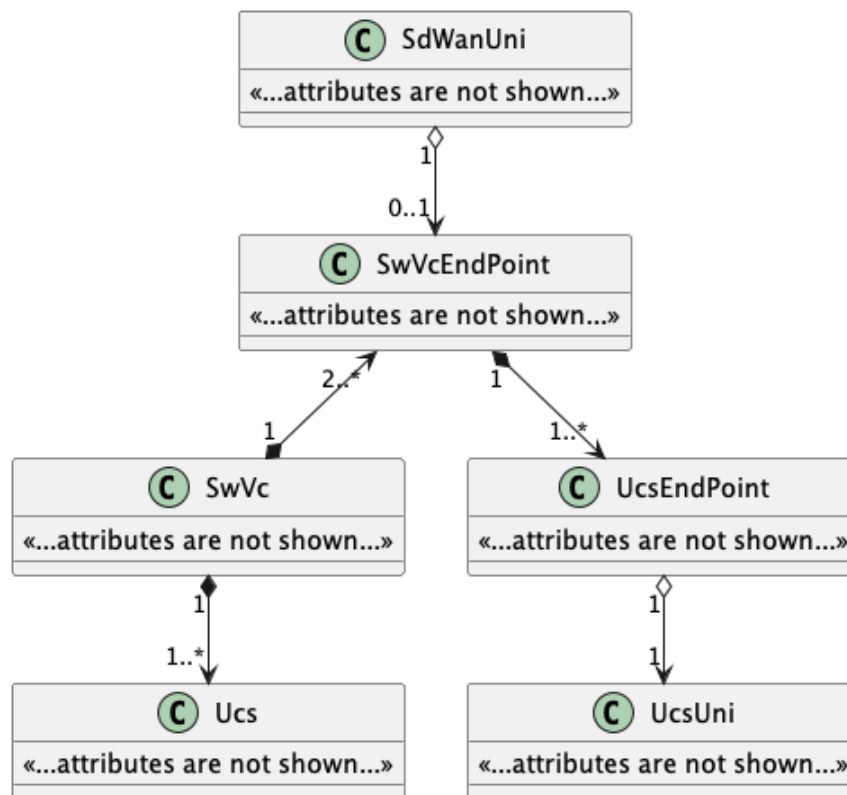


Figure 4-SD-WAN and UCS Service Model Overview

The model above uses two types of relationships: Composition and Aggregation. Aggregation represents a “part of” relationship where each object has a separate lifeline. Composition is a special type of Aggregation where parts are destroyed when the whole is destroyed. See Figure 5.

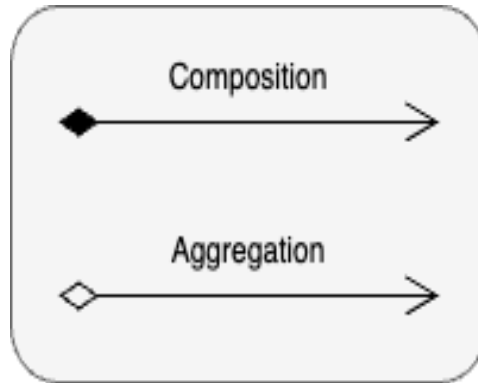


Figure 5-UML Associations

9 Data Model Design Principles and Assumptions

A Service Attribute for a Service can have a value that is a simple datatype such as an integer or string, a list of simple datatypes, a value that is an object with multiple properties, or a composition of objects. Within this document each simple value (integer, string, Boolean, etc.) is referred to as a Service-Specific Attribute. A Service-Specific Attribute could be a Service Attribute (in the case where the Service Attribute itself has a simple type) or it could be a parameter within a Service Attribute (if the Service Attribute is a structured object or a composition of such objects). The classification for each Service-Specific Attribute may be different across Service Function, Service Action, and Service Offering.

The SD-WAN Service data model supports both INSTALL and CHANGE actions for Service Order for SD-WAN VC, SD-WAN UNI, and SD-WAN VC End Point. The SD-WAN Service data model supports the RETRIEVE action for Inventory for all Service Order components.

The location and physical layer of a SD-WAN UNI cannot be changed once it is ordered; instead, this is handled as an installation (SD-WAN UNI at new location) and disconnect (SD-WAN UNI at previous location), as there is often a requirement for a smooth transition with minimum downtime.

10 Data Model for SD-WAN Services

The data model for the SD-WAN Service configuration are expressed as a set of JSON schemas based on JSON schema draft 7 [1] and encoded in YAML. These schemas accompany this document. This section explains the organization and structure of these schemas.

10.1 Organization and Structure of the Schemas

The schemas are organized into a file structure as shown in Figure 6-Schema Files Organization.

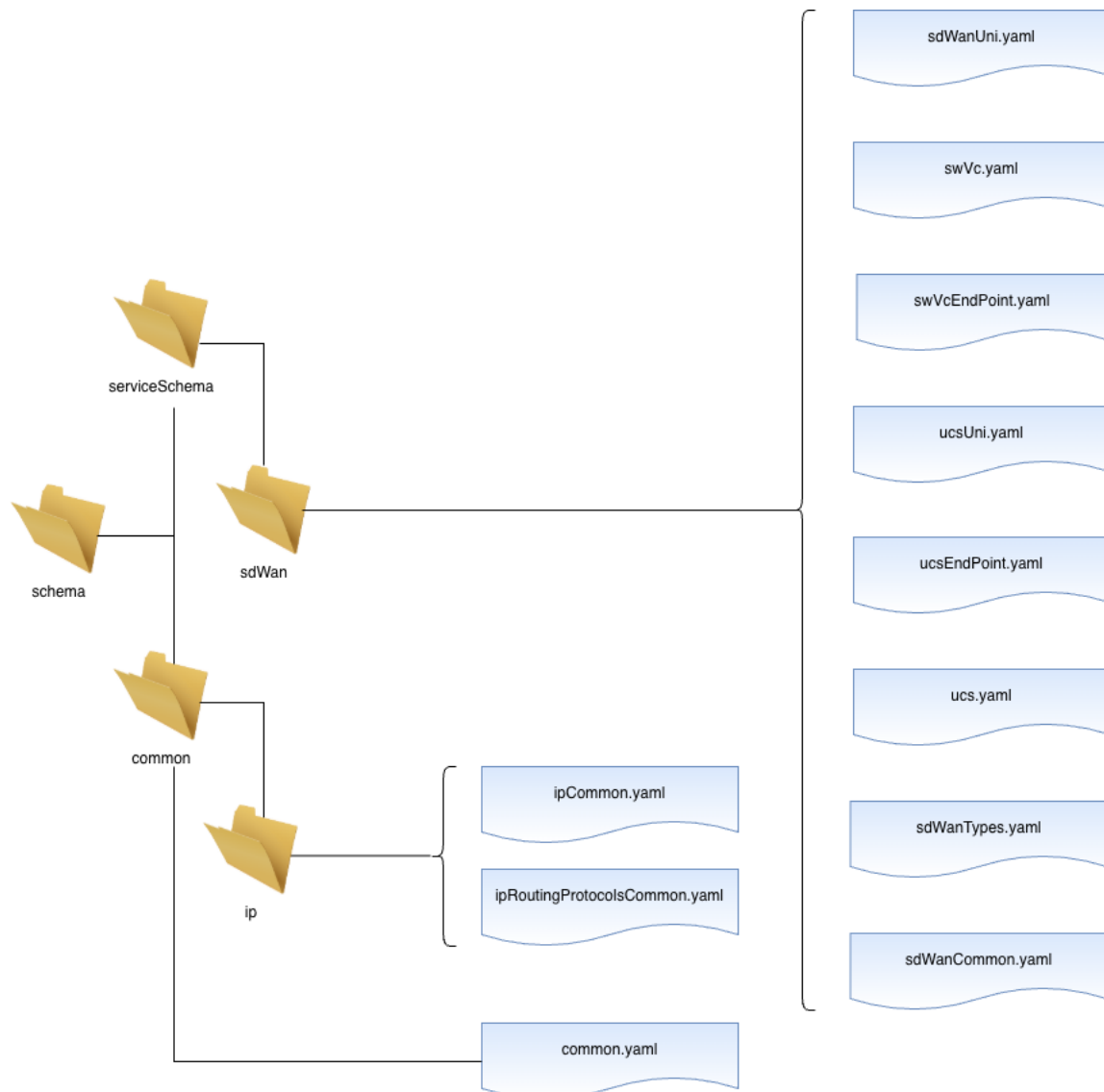


Figure 6-Schema Files Organization

SD-WAN schemas are put into two directories, sdWan and common. In the case of the SD-WAN – all components are manageable as a single service. SwVc, SdWanUni, SwVcEndPoint, Ucs, UcsEndPoint and UcsUni are considered separate SD-WAN service order items thus ordering a full configuration would require providing all off these resources in a Service Order request.

10.2 Naming Conventions

In the schemas, class and type names are UpperCamelCase and Service Attribute/property names are lowerCamelCase.

11 Relationships Between Entities

This section describes the constraints and relationships between the primary Service Order Items (SD-WAN Virtual Connection, SD-WAN Virtual Connection End Point and SD-WAN UNI) for SD-WAN services. Secondly, additional Service Order Items include Underlay Connectivity Service, Underlay Connectivity Service End Point, and Underlay UNI, the combination of which provide the Underlay Connectivity Service(s) for an SD-WAN Services.

The SD-WAN Service is associated with exactly one SWVC. The SWVC is associated with two or more SD-WAN Virtual Connection End Points. The SD-WAN UNI is associated with an SD-WAN Virtual Connection End Point. Figure 7 illustrates the Service-agnostic Service Order with several Service Order Items and their Service-specific relationship to a SD-WAN Service.

A Service Order is composed of one or more Service Order Items. This is supported in the service-agnostic part of the Service Order API. The service-specific payload (SD-WAN Services) is where the main components are supported as part of SD-WAN Service schemas. Each Service Order Item is then associated with a service-specific orderable component (i.e., SwVc) which is within the payload as shown in Figure 7.

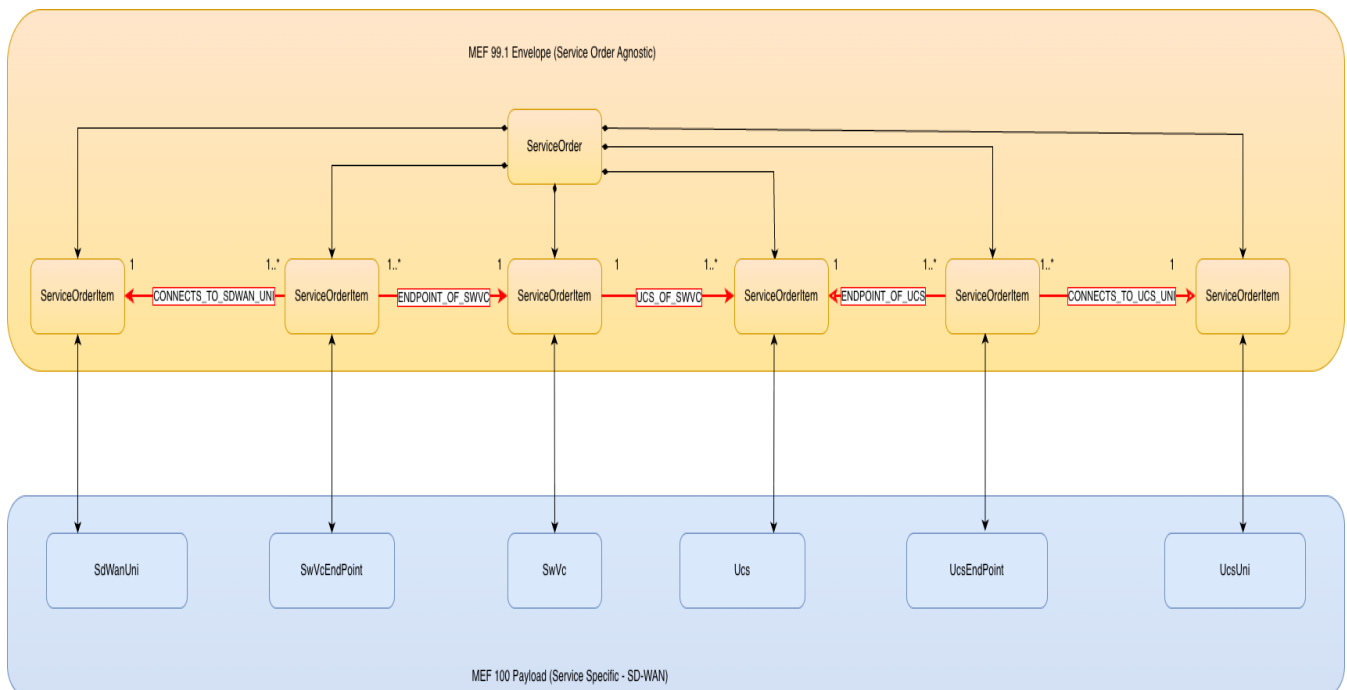


Figure 7-SD-WAN and UCS Service Order API Associations

The relationship between separately managed Service Order Items is captured in the service-agnostic part of the Order API. The values in the Relationship Type column in the table below are used in the *relationshipType* field of the *ServiceRelationship*, and *OrderItemRelationship* types.

Source Service Resource	Relationship Type	Cardinality	Target Service Resource
SwVcEndPoint	ENDPOINT_OF_SWVC	1..*	SwVc
SwVcEndPoint	CONNECTS_TO_SDWAN_UNI	1..*	SdWanUni
SwVc	UCS_OF_SWVC	1..*	Ucs
UcsEndPoint	ENDPOINT_OF_UCS	1..*	Ucs
UcsEndPoint	CONNECTS_TO_UCS_UNI	1..*	UcsUni

Table 4-SD-WAN Service and UCS Relationship Type

[R1] For an SD-WAN Service, the Relationship Type field of the Service Relationship Order Item Relationship types **MUST** contain the value shown in the Relationship Type column in Table 4.

[R2] For Order, the relationship to a SD-WAN UNI **MUST** be specified for every INSTALL of, or CHANGE to, an SD-WAN service.

Service	Place Relationship Type	INSTALL	CHANGE
SdWanUni	INSTALL_LOCATION	Mandatory	Mandatory

Table 5-Place Relationship Type

[R3] For SD-WAN UNI products, the Type field (*type*) of the Related Place (*RelatedPlaceRefOrValue*) type **MUST** contain the INSTALL_LOCATION value shown in the Place Relationship Type column in Table 5.

[R4] For Order, the Related Place (*RelatedPlaceRefOrValue*) **MUST** be specified for every INSTALL of, or CHANGE to, an SD-WAN UNI.

[R5] For a CHANGE to a SD-WAN UNI product the Related Place **MUST NOT** be changed from the value present in the Service Inventory.

The SD-WAN service specification defines several Service Order Items that are defined in the payload with associations defined in the envelope part of the Service Order API (Mplify 99.1). The set of associations and corresponding main resources (objects) are illustrated in Figure 8.

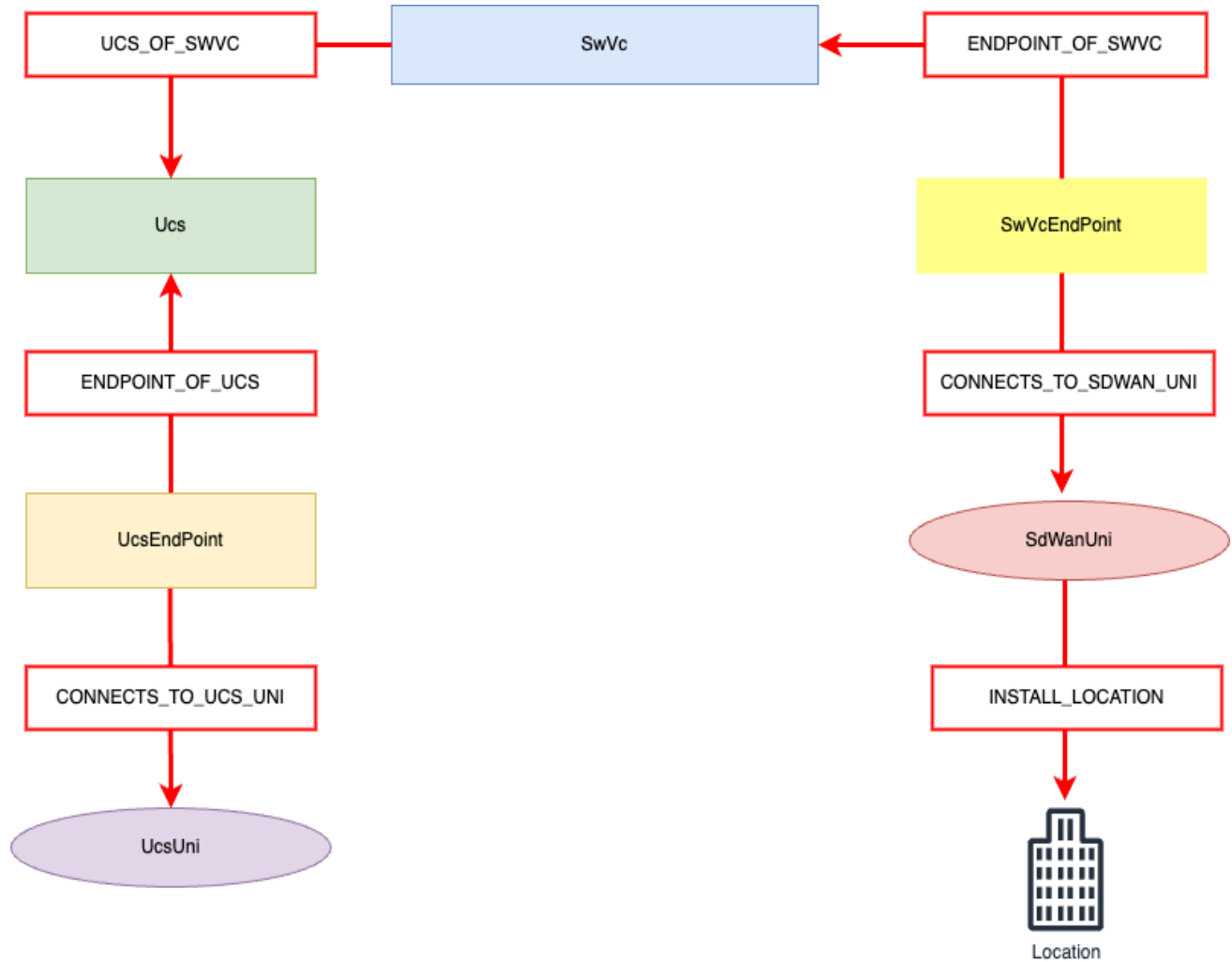


Figure 8-SD-WAN and UCS Service Relationships

12 SD-WAN Service Data Model

A SD-WAN Service is an SD-WAN Service provided to an end user (the Subscriber) by a Service Provider. There is no restriction on the type of organization that can act as a Subscriber; for example, a Subscriber can be an enterprise, a mobile operator, an IT system integrator, a government department, etc. At its most basic, a Subscriber SD-WAN Service provides connectivity for IP Packets between different parts of the Subscriber's network (usually at different physical locations) or between the Subscriber's network and an external network, such as the public Internet or a private cloud service.

The Service Attributes are listed in groups:

- SD-WAN Services:
 - SwVc
 - SwvcEndPoint
 - SdwanUni
 - Ucs
 - UcsEndPoint
 - UcsUni

12.1 SwVc

The SwVc is an association of SwVcEndPoints and the SwVc includes Service Attribute that identifies the SwVEndPoints associated by the SwVc. SwVc is a resource that represents a SD-WAN Virtual Connection. Reference MEF 70.2 [13] Section 9 SD-WAN Virtual Connection (SWVC) Service Attributes.

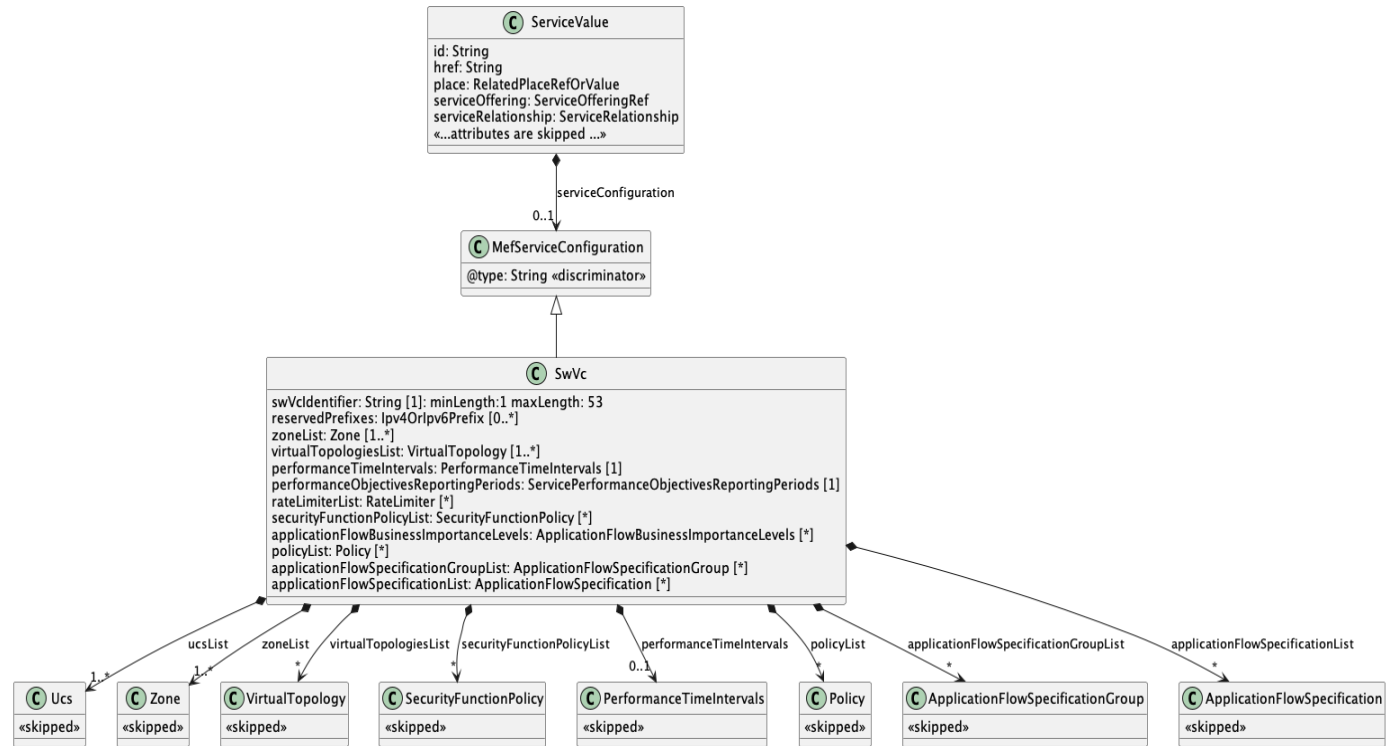


Figure 9-SwVc Service Model

Schema File Name: schema/serviceSchema/sdWan/swVc.yaml

Specified Service: SD-WAN VC

\$Id: urn:mef:iso:spec:service:swvc:v0.0.3:all

Attribute Name	Type	Multiplicity	Description
swVcIdentifier	String <i>Min length=1</i> <i>Max length=53</i> <i>Pattern=pattern:</i> <i>"[x20-xf]+"</i>	1	Identification of the SWVC for management purposes. Reference MEF 70.2, Section 9.1 SWVC Identifier Service Attribute.
reservedPrefixes	Ipv4OrIpv6Prefix	0..*	Specifies a list of IPv6 Prefixes that the Service Provider reserves for use for the SWVC within their own network or for distribution to the Subscriber via DHCP or SLAAC. Reference MEF 70.2 Section 9.4 SWVC Reserved Prefixes Service Attribute.
zoneList	Zone	1..*	A list of the Zones supported at one or more UNIs in the SD-WAN Service and Zone-wide Policies associated with the Zones. Reference MEF 70.2 Section 9.5 SWVC List of Zones Service Attribute.
virtualTopologiesList	VirtualTopology	1..*	A list of named Virtual Topologies of the form of a 3-tuple <vtName,vtType,vtEP>. Reference MEF 70.2

			Section 9.6 SWVC List of Virtual Topologies Service Attribute.
performanceTimeIntervals	PerformanceTimeIntervals	0..1	Time intervals used in the evaluation of the PERFORMANCE Policy Criterion and the Performance Metrics. Reference MEF 70.2 Section 9.7 SWVC Performance Time Intervals Service Attribute.
performanceObjectivesReportingPeriods	ServicePerformanceObjectivesReportingPeriod	0..1	Definition of reporting periods for evaluating Service Performance Objectives. Reference MEF 70.2 Section 9.8 SWVC Service Performance Objectives Reporting Periods Service Attribute.
rateLimiterList	RateLimiter	0..*	A 3-tuple, <name,commit,limit> where: * name is an Identifier String that is name of Rate Limiter. * commit is the threshold rate (bits per second) at or below which the SD-WAN Service Provider commits to deliver packets in the Bandwidth Flow with high probability under all traffic conditions. * limit is the threshold information rate (bits per second) above which the Service Provider does not deliver IP Packet in the Bandwidth Flow under any conditions. Reference MEF 70.2 Section 9.9 SWVC List of Rate Limiters Service Attribute.
securityFunctionPolicyList	SecurityFunctionPolicy	0..*	A list of the Security Policies (defined in MEF 138) that are available for use in this SWVC. Reference MEF 70.2 Section 9.10 SWVC List of Security Policies Service Attribute.
applicationFlowBusinessImportanceLevels	ApplicationFlowBusinessImportanceLevels	0..*	Specifies and ordered list of labels that can be assigned to Application Flows to indicate the relative importance of each Application Flow. Reference MEF 70.2 Section 9.11 SWVC List of Application Flow Business Importance Levels Service Attribute.
policyList	Policy	1..*	A list of the Policies that can be applied to Application Flows carried by the SWVC. Reference MEF 70.2 Section 9.10 SWVC List of Policies Service Attribute.
applicationFlowSpecificationGroupList	ApplicationFlowGroup	0..*	A list (possibly empty) of Application Flow Group names. Reference MEF 70.2 Section 9.13 SWVC List of Application Flow Groups Service Attribute.
applicationFlowSpecificationList	ApplicationFlowSpecification	1..*	Specifies the Application Flows that can be recognized by the SD-WAN Service. Reference MEF

			70.2 Section 9.14 SWVC List of Application Flow Specifications Service Attribute.
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Table 6-SwVc Service Attributes

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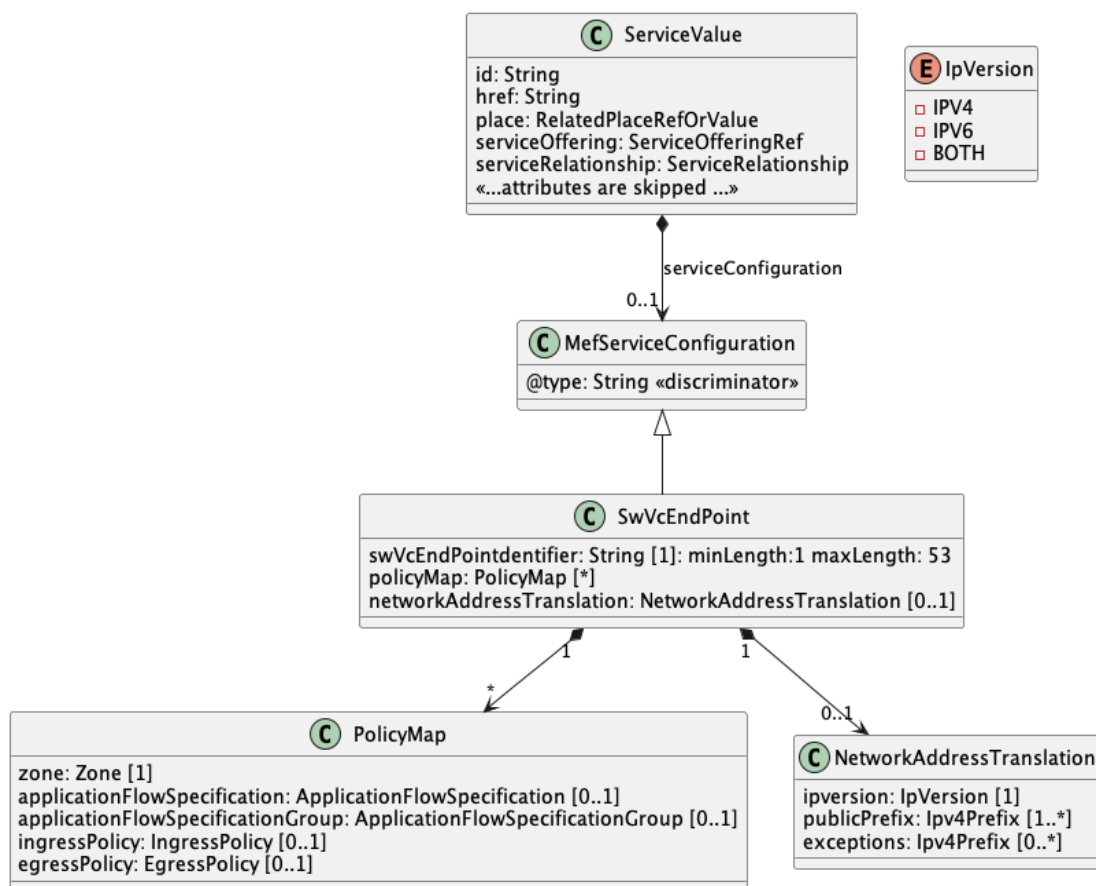
The SWVC list of SWVC End Points and SWVC List of UCSs attributes as defined in MEF 70.2[13] Section 9 SD-WAN Virtual Connection (SWVC) Service Attributes are not defined in the SD-WAN service schema. The reason is these are attributes that are represented as relationships between a SWVC and SWVC End Point and UCS respectively. Relationships are part of the Service Order envelope.

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12.2 SwVcEndPoint

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The SwVcEndPoint is the construct that represents the attachment of an SWVC to a SD-WAN UNI. The SwVcEndPoint provides a container for attributes of the SWVC that can differ at each SD-WAN UNI. Reference MEF 70.2[13] Section 10 SD-WAN Virtual Connection (SWVC) End Point.



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Figure 10-SwVcEndPoint Service Model

Schema File Name: schema/serviceSchema/sdWan/swVcEndPoint.yaml Specified Service: SD-WAN VC End Point \$Id: urn:mef:lso:spec:service:swvc-end-point:v0.0.3:all			
Attribute Name	Type	Multiplicity	Description
swVcEndPointIdentifier	String <i>Min length=1</i> <i>Max length=53</i> <i>Pattern=pattern:</i> <i>"[x20-x7F]+"</i>	1	The value of the SWVC End Point Identifier Service Attribute is a string that is used to allow the Subscriber and Service Provider to uniquely identify the association of the SWVC with a UNI. Reference MEF 70.2 Section 10.1 SWVC End Point Identifier Service Attribute.
policyMap	PolicyMap	0..*	Maps Policies to Application Flows and Application Flow Groups. Reference MEF 70 Section 9.3 SWVC End Point Policy Map.
networkAddressTranslation	NetworkAddressTranslation	0..1	When Ingress IP Packets are destined to the Internet via Internet Breakout, the Subscriber might require the SD-WAN Service Provider to perform Network Address Translation (NAT), translating the Subscriber's IP addresses into public IP addresses and vice versa. Reference MEF 70.2 Section 10.5 SWVC End Point Network Address Translation Service Attribute.

Table 7-SwVcEndPoint Service Attributes

The SWVC End Point Associated UNI and SWVC End Point List of UCS End Points attributes defined in MEF 70.2[13] Section 10 SD-WAN Virtual Connection (SWVC) End Point Service Attributes are not defined in the SD-WAN service schema. The reason is these are attributes that are represented as relationships between SWVC UNI and UCS End Points respectively. Relationships are part of the Service Order envelope.

12.3 SdWanUni

An SD-WAN User Network Interface, or SD-WAN UNI, is the demarcation point between the responsibility of the Service Provider and the responsibility of the Subscriber. The SD-WAN UNI is on the boundary between the Subscriber Network and the Service Provider Network. The basic unit of transport at the SD-WAN UNI is an IP Packet. Reference MEF 70.2[13] Section 7.4 SD-WAN UNI.

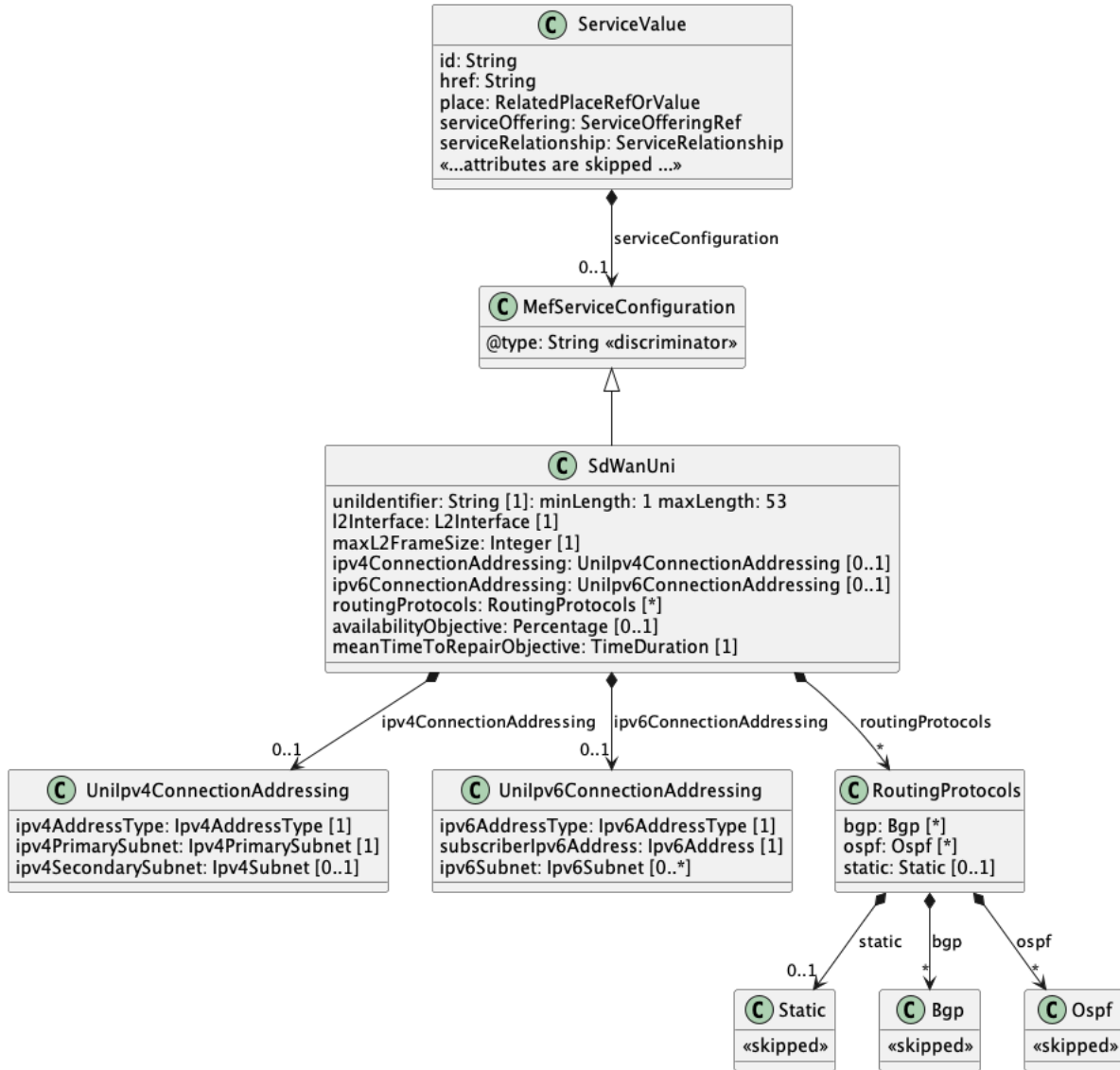


Figure 11-SdWanUni Model

Schema File Name: schema/serviceSchema/sdWan/sdWanUni.yaml			
Specified Service: SD-WAN UNI			
\$id: urn:mef:iso:spec:service:sd-wan-uni:v0.0.3:all			
Attribute Name	Type	Multiplicity	Description
unIdentifier	String Min length=1 Max length=53 Pattern=pattern: "[\x20-\x7F]+"	1	Identification of the UNI for management purposes. Reference MEF 70.2, Section 11.1 SD-WAN UNI Identifier Service Attribute.
l2Interface	L2Interface	1	Describes the underlying L2 technology for the UNI. Reference MEF 70.2, Section 11.2 SD-WAN UNI L2 Interface Service Attribute.

maxL2FrameSize	Integer	1	Specifies the maximum length L2 frame that is accepted by the Service Provider. Reference MEF 70.2, Section 11.3 SD-WAN UNI Maximum L2 Frame Size Service Attribute.
ipv4ConnectionAddressing	Unilpv4ConnectionAddressing	0..1	Specifies how IPv4 addresses are allocated to the devices on the Subscriber side for the UNI. This attribute is omitted if there is no IPv4 address (i.e., if the Service Attribute value is None). Reference MEF 70.2, Section 11.4 SD-WAN UNI IPv4 Connection Addressing Service Attribute.
ipv6ConnectionAddressing	Unilpv6ConnectionAddressing	0..1	Specifies how IPv6 addresses are allocated to the devices connected to the UNI. This attribute is omitted if there is no IPv6 address (i.e., if the Service Attribute value is None). Reference MEF 70.2 Section 11.5 SD-WAN UNI IPv6 Connection Addressing Service Attribute.
routingProtocols	RoutingProtocols	0..*	List of Routing Protocols used across the UNI. Reference MEF 70.2 Section 11.6 SD-WAN UNI Routing Protocols Service Attribute.
availabilityObjective	Percentage	0..1	UNI Availability. Reference MEF 70.2 Section 11.7 SD-WAN UNI Availability Objective Service Attribute.
meanTimeToRepairObjective	TimeDuration	0..1	Mean Time to Repair Objective. Reference MEF 70.2 Section 11.8 SD-WAN UNI Mean Time To Repair Objective Service Attribute.

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Table 8-SdWanUni Service Attributes

13 Underlay Connectivity Service Data Model

This section details the associated set of objects and data types that define the Underlay Connectivity Service model. Underlay Connectivity Services are network services independent of the SD-WAN Service and can have many “characteristics” or “attributes” that define their configuration and behavior.

The UCS and UCS End Point Service Attributes defined in this model are in addition to those attributes and include only those attributes necessary to define UCS external interfaces and behavior to the extent necessary to implement SD-WAN Policies.

Attributes that are internal to the UCS itself, such as those defining the associations between the UCS and UCS End Points or between the UCS End Point and UCS UNI are part of the UCS Service Agreement described earlier in this paragraph and are not included as Service Attributes for the SD-WAN Service.

13.1 Ucs

Underlay Connectivity Services are network services independent of the SD-WAN Service and can have many 'characteristics' or 'attributes' that define their configuration and behavior. Reference MEF 70.2 Section 12 UCS Service Attributes.

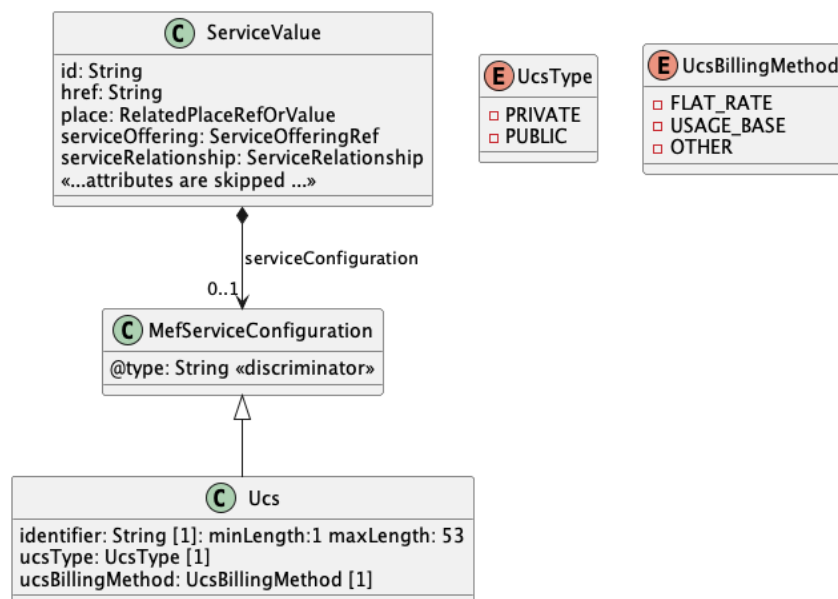


Figure 12-Ucs Model

Schema File Name: schema/serviceSchema/sdWan/ucs.yaml Specified Service: UCS \$id: urn:mef:iso:spec:service:ucs:v0.0.3:all			
Attribute Name	Type	Multiplicity	Description
ucsIdentifier	String <i>Min length=1</i> <i>Max length=53</i> <i>Pattern=pattern:</i> <i>"[x20-x7F]+"</i>	1	The value of the UCS Identifier Service Attribute is a string that is used to allow the Subscriber and Service Provider to uniquely identify an Underlay Connectivity Service. Reference MEF 70.2 Section 12.1 UCS Identifier Service Attribute.
ucsType	UcsType	1	The value of the UCS Type Service Attribute indicates whether the UCS is a Public UCS (i.e., an Internet Access Service) or a Private UCS. The possible values are Public or Private. Reference MEF 70.2 Section 12.2 UCS Type Service Attribute.
ucsBillingMethod	UcsBillingMethod	1	Indicates how access to the Underlay Connectivity Service is billed. The allow values are FLAT_RATE, USAGE_BASED and OTHER. Reference MEF 70.2 Section 12.3 UCS Billing Method Service Attribute.

Table 9-Ucs Attributes

13.2 UcsUni

Access to an Underlay Connectivity Service is provided at the SD-WAN Edge via the UCS UNI.

Reference MEF 70.2 Section 13 UCS UNI Service Attribute.

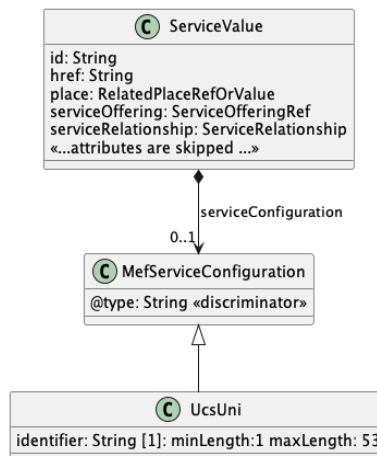


Figure 13-UcsUni Model

Schema File Name: schema/serviceSchema/sdWan/ucsUni.yaml Specified Service: UCS UNI \$id: urn:mef:iso:spec:service:ucs-uni:v0.0.3:all			
Attribute Name	Type	Multiplicity	Description
identifier	String Min length=1 Max length=53 Pattern=pattern: "[\x20-\x7F]+"	1	The value of the UCS UNI Identifier Service Attribute is a string that is used to allow the Subscriber and Service Provider to uniquely identify an Underlay Connectivity Service UNI. Reference MEF 70.2 Section 13 UCS UNI Service Attributes.

Table 10-UcsUni Attributes

13.3 UcsEndPoint

A UCS is connected to a UCS UNI by a UCS End Point. Some service architectures support multiple UCSs at the same UCS UNI. The UCS End Point captures Service Attributes at the UCS UNI whose value can differ between UCSs. Reference MEF 70.2 Section 14 UCS End Point Service Attributes.

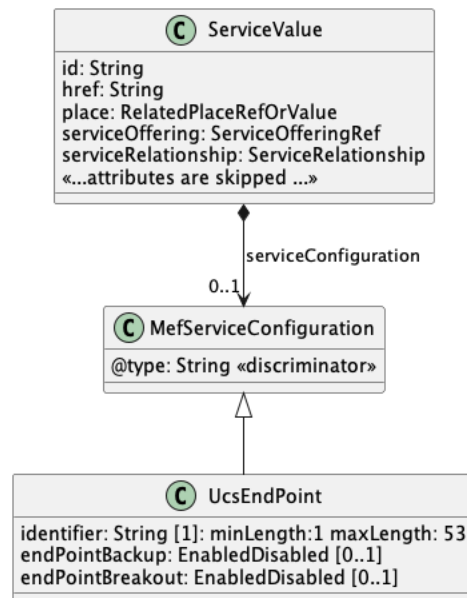


Figure 14-UcsEndPoint Model

Schema File Name: schema/serviceSchema/sdWan/ucsUniEndPoint.yaml Specified Service: UCS UNI End Point \$id: urn:mef:iso:spec:service:ucs-uni-end-point:v0.0.3:all			
Attribute Name	Type	Multiplicity	Description
identifier	String Min length=1 Max length=53	1	Identification of the Underlay Connectivity Service End Point. Reference MEF 70.2 Section 14.1

	<i>Pattern=pattern:</i> " <i>[\x20-\x7F]+</i> "		UCS End Point Identifier Service Attribute.
endPointBackup	EnabledDisabled	1	Indicates whether the UCS should be treated as a Primary underlay or a Backup underlay at this UCS End Point. Reference MEF 70.2 Section 14.2 UCS End Point Backup Service Attribute.
endPointBreakout	EnabledDisabled	1	Indicates whether the SD-WAN Service can "break out" to the underlying UCS Service from this End Point. Reference MEF 70.2 Section 14.3 UCS End Point Breakout Service Attribute.

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Table 11-UcsEndPoint Attributes

14 Common Resources and Types

This section is structured like the previous section but focuses on common resources and types used by the Service Attributes. Most of these are structured to support a variety of Mplify Services. This section details the data types and enumerations that are used by the many Mplify Service models.

14.1 EnabledDisabled

Enumeration used to indicate Enabled/Disabled state of an attribute.

Schema File Name: schema/common/common.yaml

Contains Enumeration Literals:

- ENABLED
- DISABLED

14.2 FourOctetInteger

A four-octet value range integer 0-4294967295.

Schema File Name: schema/common/common.yaml

type: integer

format: int64

minimum: 0

maximum: 4294967295

14.3 InformationRate

A value and a unit of measure that specifies an Information Rate.

Schema File Name: schema/common/common.yaml			
Attribute Name	Type	Multiplicity	Description
irValue	Number Minimum: 0	1	The value of the information rate. For example, if the rate is 70 kbps, 70 is the value.
irUnits	InformationRateUnits	1	The unit of measure for the Information Rate. For example, if the Information Rate is 70KBPS this element is KBPS. Note that the values are decimal values. 1 KBPS is 1000 bits per second and 1MBPS is 1,000,000 bits per second.

Table 12-InformationRate Attributes

14.4 InformationRateUnits

Enumeration representing information rate units.

Schema File Name: schema/common/common.yaml

Contains Enumeration Literals:

- BPS:
 - Bits per second.
- Kbps:
 - Kilobits per second.
- MBPS:
 - Megabits per second.
- GBPS:
 - Gigabits per second.
- TBPS:
 - Terabits per second.
- PBPS:
 - Petabits per second.
- EBPS:
 - Exabits per second.
- ZBPS:
 - Zettabits per second.
- YBPS:
 - Yottabits per second.

14.5 Percentage

This is a number of percent – a number between 0 and 100.

Schema File Name: schema/common/common.yaml

type: number

minimum: 0

maximum: 100

14.6 TimeDuration

This class is used to describe durations expressed as a 2-tuple, (value, units). The units from nanoseconds to years.

Schema File Name: schema/common/common.yaml			
Attribute Name	Type	Multiplicity	Description
timeDurationValue	Integer	1	The value of the duration. For example, if the duration is 20 ms, this element is 20.
timeDurationUnits	TimeDurationUnits	1	The unit of measure in the duration. For example, if an interval is 2ms, this element is MS.

Table 13-TimeDuration Attributes

14.7 TimeDurationUnits

Enum listing the units of measure for a Time Duration.

Schema File Name: schema/common/common.yaml

Contains Enumeration Literals:

- NS:
 - Nanoseconds
- US:
 - Microseconds
- MS
 - Milliseconds
- SEC
 - Seconds
- MIN
 - Minutes
- HOUR
 - Hour
- DAY
- WEEK
- MONTH
- YEAR

15 IP Common Resources and Types

This section is structure like the previous section but focuses on common resources and types used by the Service Attributes. Most of these are structured to support a variety of IP-based Services. This section details the data types and enumerations that are used by the SD-WAN Service model.

15.1 PeeringAddress

Peering Addresses. Connection Addresses, or Loopbacks plus a list of pairs of IP addresses. Reference MEF 61.1 Section 12.7.3 BGP and MEF 70.2 Section 11.6.2 BGP.

Schema File Name: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
subscriberAndSpLoopbackAddresses	SubscriberAndSpLoopbackAddresses	0..*	A list of pairs of IP addresses, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses. <i>NOTE: SD-WAN will have a single pair of IP addresses.</i> <i>Review.</i>
peeringAddressType	PeeringAddressType	0..1	SD-WAN unlike IP (MEF 61.1) will only use LOOPBACKS address with a single BGP Session. <i>NOTE: Review</i>

Table 14-PeeringAddress Attributes

15.2 PeeringAddressType

If the Peering Addresses parameter is CONNECTION_ADDRESSES, a separate BGP peering session is established over each UNI Access Link, using the primary IPv4 addresses in the UNI Access Link IPv4 Connection Addressing Service Attribute (section 13.4) or the first IPv6 addresses in the UNI Access Link IPv6 Connection Addressing Service Attribute (section 13.5), as indicated by the Connection Address Family parameter. If the Peering Addresses parameter is LOOPBACKS, a list of pairs of IP addresses is additionally specified, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses.

File: /sdWan/common/ipCommon.yaml

- CONNECTION_ADDRESSES
- LOOPBACKS

15.3 IP Addressing Model

The following section provides schema to support IPv4 and IPv6 addresses for SD-WAN service. Note that the API schema leverages the OAS embedded *ipv4* and *ipv6* formats and uses them to specify the *Ipv4Address* and *Ipv6Address* data types that are used when an address value is provided.



Figure 15-Ipv4Address Model

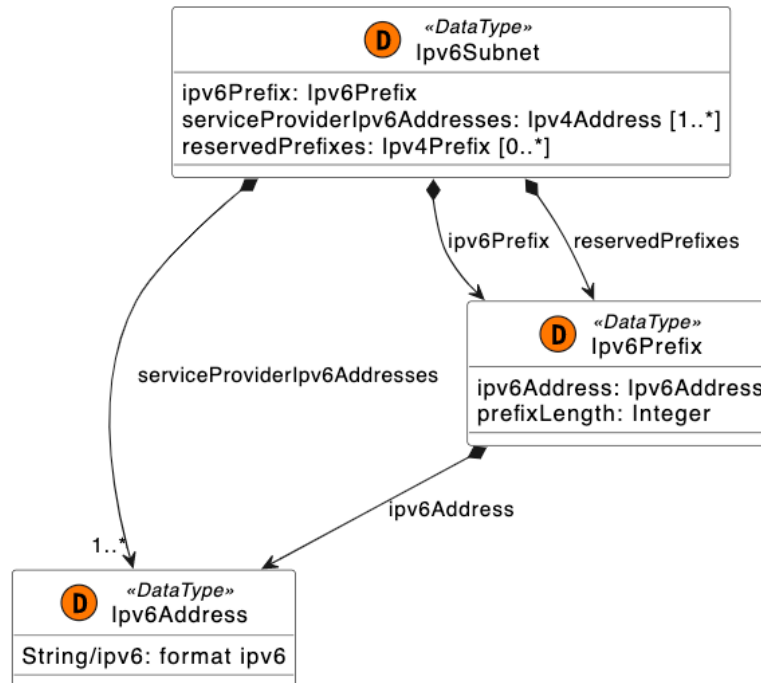


Figure 16-Ipv6Address Model

15.3.1 Any

File: /sdWan/common/ipCommon.yaml

Used to specify ANY for IPv4 or IPv6 address.

- ANY

15.3.2 Dscp

File: /sdWan/common/ipCommon.yaml

Differentiated Services Code Point.

type: integer

maximum: 63

minimum: 0

15.3.3 Ipv4Address

Data type representing IPv4 address.

File: schema/common/ipCommon.yaml

- type: String
- format: ipv4

15.3.4 Ipv4AddressOrAny

Data type which allows an IPv4 address or any IPv4 address to be specified.

Schema File Name: schema/service/sdWan/sdWanCommon.yaml

minProperties: 1

maxProperties: 1

Attribute Name	Type	Multiplicity	Description
ipv4Address	Ipv4Address	1	IPv4 Address.
ipv4Any	String	1	Any IPv4 address.

Table 15-Ipv4AddressOrAny Attributes

15.3.5 Ipv4ConnectionAddressing

A data type representing how IPv4 addresses are allocated to the devices on the Subscriber side.

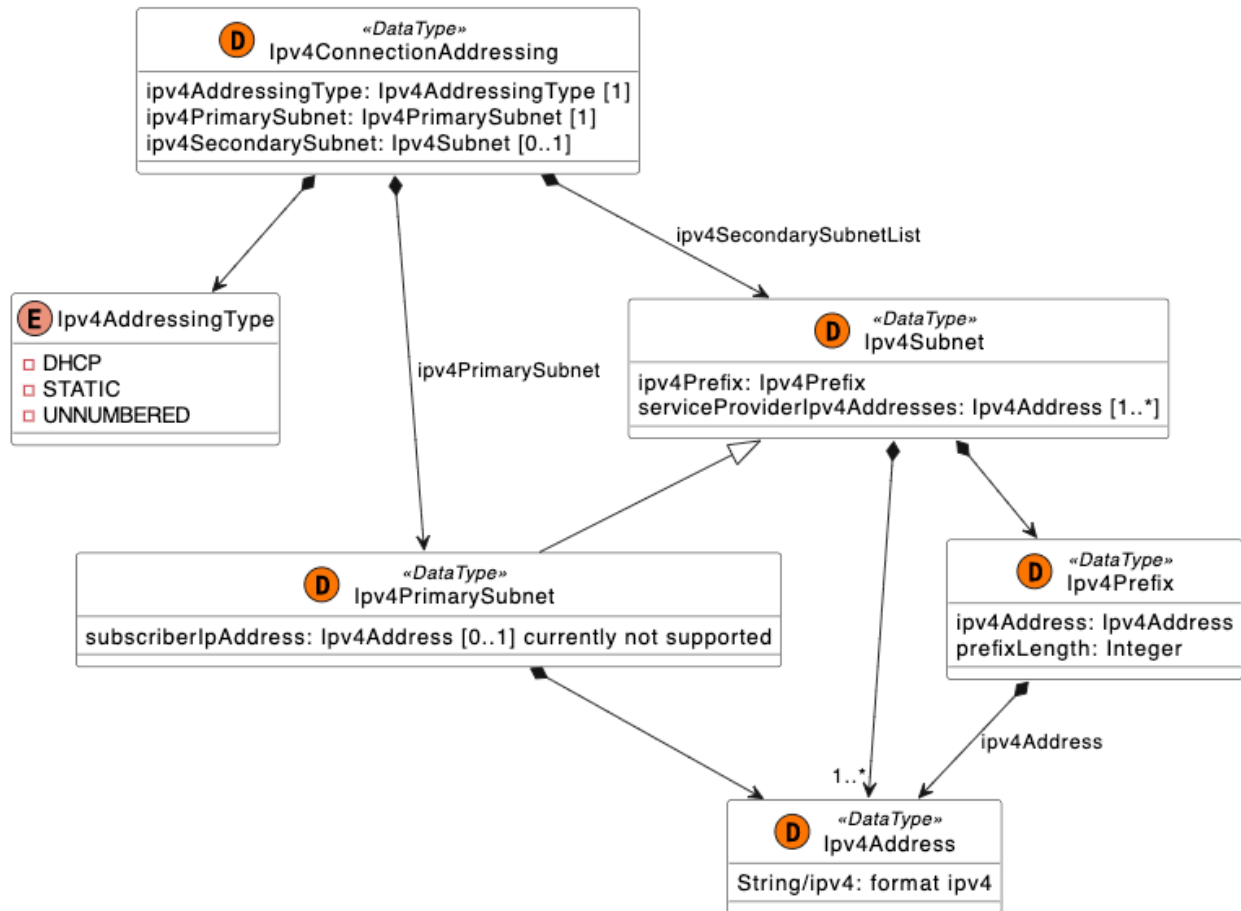


Figure 17-Ipv4ConnectionAddressing Model

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4AddressingType	Ipv4AddressingType	1	IPv4 address type.
ipv4PrimarySubnet	Ipv4PrimarySubnet	1	Primary IPv4 Subnet.
Ipv4SecondarySubnet	Ipv4Subnet	0..*	Secondary IPv4 subnet list.

Table 16-Ipv4ConnectionAddressing Attributes

15.3.6 Ipv4AddressingType

Enumeration representing IPv4 Address Types.

File: /sdWan/common/ipCommon.yaml

- DHCP:

- Dynamic Host Configuration Protocol (DHCP) is used the Subscriber devices to request IPv4 addresses in each subnet from the SP or Operator.
- **STATIC:**
 - IPv4 addresses in each IPv4 subnet are statically assigned to the SP or Operator and to the Subscriber.
- **UNNUMBERED:**
 - The SP or Operator and the Subscriber each assigned an IPv4 address from their own address pools independently. These addresses can be on different subnets, and so an interface-based routing protocol is needed to ensure reachability.

15.3.7 Ipv4Prefix

Data type representing IPv4 address prefix and mask length between 0 and 31 bits.

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
prefixLength	Integer	1	IPv4 address prefix. Length 0-31.
Ipv4Address	Ipv4Address	1	IPv4 address.

Table 17-Ipv4Prefix Attributes

15.3.8 Ipv4Subnet

Data type representing an IPv4 subnet logical partition of an IP network. Included is list of Service Provider IPv4 addresses.

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4Prefix	Ipv4Prefix	1	IPv4 address prefix (IPv4 address prefix and mask length between 0 and 31 in bits).
serviceProviderIpv4Addresses	Ipv4Address	1..*	List of Service Provider IPv4 addresses. [R25] If IPv4 Connection Addressing is DHCP, the UNI Access Link IPv4 Connection Addressing Primary Subnet parameter MUST contain only a single Service Provider IPv4 Address. Reference MEF 69.1[1] Section 9.4 Subscriber

			Internet Access Service: UNI Access Link Requirements
reservedPrefixes	Ipv4Prefix	0..*	List of IPv4Prefixes, possibly empty.

Table 18-Ipv4Subnet Attributes

15.3.9 Ipv4PrimarySubnet

Used in context of Primary subnet. It adds the subscriberIpv4Address attribute to the Ipv4Subnet.

File: /sdWan/common/ipCommon.yaml Inherits from: Ipv4Subnet			
Attribute Name	Type	Multiplicity	Description
subscriberIpv4Address	Ipv4Address	0..1	Subscriber IPv4 Address.

Table 19-Ipv4PrimarySubnet Attributes

15.3.10 Ipv6Address

Data type representing IPv6 address.

File: schema/common/ipCommon.yaml

- type: String
- format: ipv6

15.3.11 Ipv6AddressOrAny

Data type which allows an IPv6 address or any IPv6 address to be specified.

Schema File Name: schema/service/sdWan/sdWanCommon.yaml minProperties: 1 maxProperties: 1			
Attribute Name	Type	Multiplicity	Description
Ipv6Address	Ipv6Address	1	IPv6 Address.
Ipv6Any	String	1	Any IPv6 address.

Table 20-Ipv6AddressOrAny Attributes

15.3.12 Ipv6AddressingType

Enumeration representing IPv6 Address Types.

843 File: /sdWan/common/ipCommon.yaml

- 844 • DHCP:
- 845 ○ Dynamic Host Configuration Protocol (DHCP) is used the Subscriber devices to
- 846 request IPv4 addresses in each subnet from the SP.
- 847 • SLAAC:
- 848 ○ Stateless Address Autoconfiguration is used by the Subscriber devices to create
- 849 unique IPv6 global addresses within an IP prefix advertised by the SP or Operator
- 850 as described in RFC 4862 [7].
- 851 • STATIC:
- 852 ○ IPv4 addresses in each IPv4 subnet are statically assigned to the
- 853 SP or Operator and to the Subscriber.
- 854 • LL_ONLY:
- 855 ○ If the value is LL-only, these are only IPv6 addressed used on the UNI.

856 **15.3.13 Ipv6ConnectionAddressing**

857 A data type representing how IPv6 addresses are allocated to the devices on the Subscriber side.

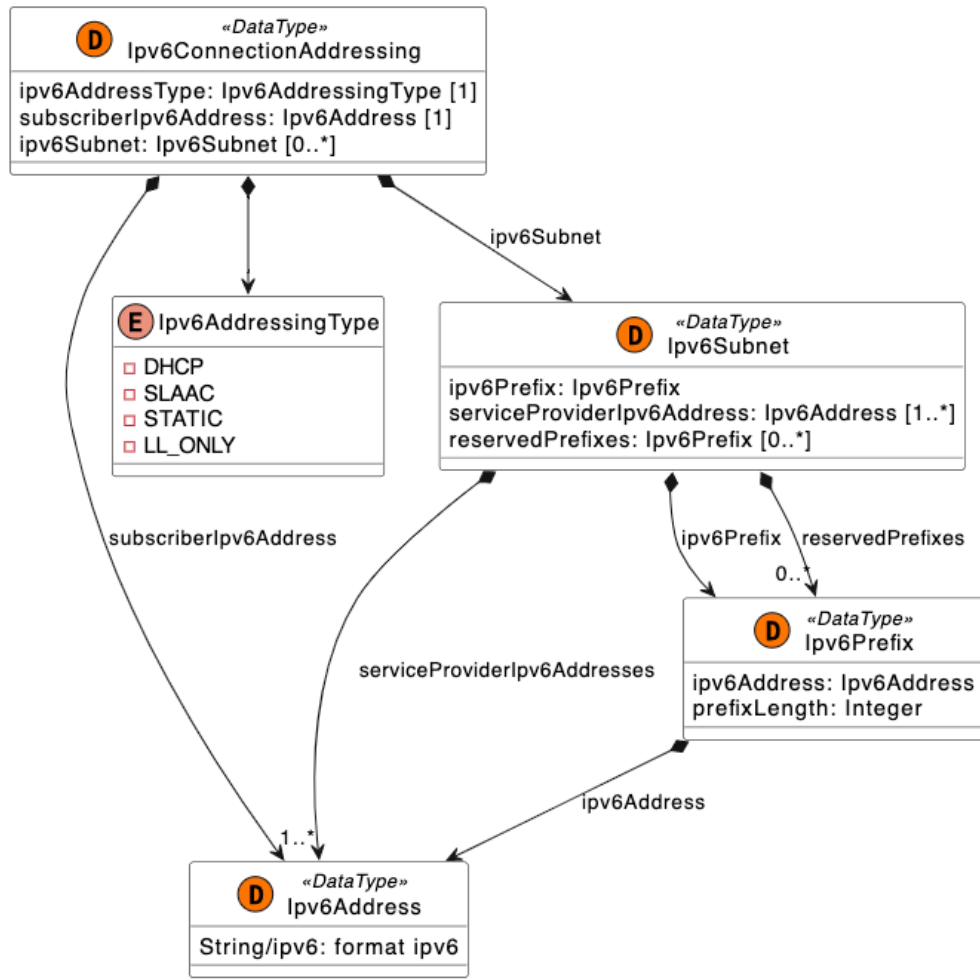


Figure 18-IPv6ConnectionAddressing Model

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv6AddressingType	IPv6AddressingType	1	IPv6 address type.
subscriberIPv6Address	IPv6Address	1	Subscriber IPv6 address.
ipv6Subnet	IPv6Subnet	0..*	IPv6 subnet list.

Table 21-IPv6ConnectionAddressing Attributes

15.3.14 IPv6AddressingType

Enumeration representing IPv6 Address Types.

File: /sdWan/common/ipCommon.yaml

- DHCP:
 - Dynamic Host Configuration Protocol (DHCP) is used the Subscriber devices to request IPv4 addresses in each subnet from the SP or Operator.
- SLAAC:
 - Stateless Address Autoconfiguration (SLAAC) is used by the Subscriber devices to create unique IPv6 global addresses in each IP Prefix advertised by SP as described in RFC 4862[7].
- STATIC:
 - IPv6 addresses in each IPv6 subnet are statically assigned to the SP and to the Subscriber.
- LL_ONLY:
 - If the value if LL-only, these are only IPv6 addresses used.

15.3.15 Ipv6Prefix

Data type representing IPv6 address prefix and mask length between 0 and 127 bits.

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
prefixLength	Integer	0..1	IPv6 address prefix. Length 0-127.
ipv6Address	Ipv6Address	1	IPv6 address.

Table 22-Ipv6Prefix Attributes

15.3.16 Ipv6Subnet

Data type representing an IPv6 subnet logical partition of an IP network. Included is list of Service Provider IPv6 addresses.

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv6Prefix	Ipv6Prefix	1	IPv6 address prefix (IPv6 address prefix and mask length between 0 and 127 in bits).
serviceProviderIpv6Addresses	Ipv6Address	1..*	List of Service Provider IPv6 addresses.
reservedPrefixes	Ipv6Prefix	0..*	List of IPv6Prefixes, possibly empty.

Table 23-Ipv6Subnet Attributes

15.3.17 Ipv4OrIpv6Prefix

Data type representing an IPv4 or IPv6 prefix. Includes subnet address and prefix length.

File: /sdWan/common/ipCommon.yaml			
one of:			
-required: [ipv4Prefix]			
-required: [ipv6Prefix]			
Attribute Name	Type	Multiplicity	Description
ipv6Prefix	Ipv6Prefix	0..1	IPv6 prefix.
ipv4Prefix	Ipv4Prefix	0..1	IPv4 prefix.

Table 24-Ipv4OrIpv6Prefix Attributes

15.3.18 Ipv4OrIpv6Address

Data type representing IPv4 or IPv6 address.

File: /sdWan/common/ipCommon.yaml

type: string

oneOf:

- format: ipv4
- format: ipv6

15.4 NetworkAddressTranslation

When Ingress IP Packets are destined to the Internet via Internet Breakout, the Subscriber might require the SD-WAN Service Provider to perform Network Address Translation (NAT), translating the Subscriber's IP addresses into public IP addresses and vice versa.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipversion	Enum: - IPV4	1	An enumeration that indicates the type(s) of Subscriber addresses that are translated.

	- IPv6 - BOTH		
publicPrefix	Ipv4Prefix	1..*	Public-assigned IPv4 Prefix representing the public address or addresses that the Subscriber's IP addresses are translated to (and from).
exceptions	Ipv4Prefix	0..*	A list (possibly empty) of IP prefixes that are not subject to translation.

Table 25-NetworkAddressTranslation Attributes

15.5 Ospf Model

When an entry in the UNI Routing Protocols is for OSPF, OSPF as specified in RFC 2328 [3] (for IPv4) and/or RFC 5340 [8] (for IPv6) is used across each UNI Access Link (IP) and UNI (for SD-WAN) to exchange routing information. Reference MEF 61.1 [12] Section 12.7.2 OSPF and MEF 70.2 [13] Section 11.6.3 OSPF.

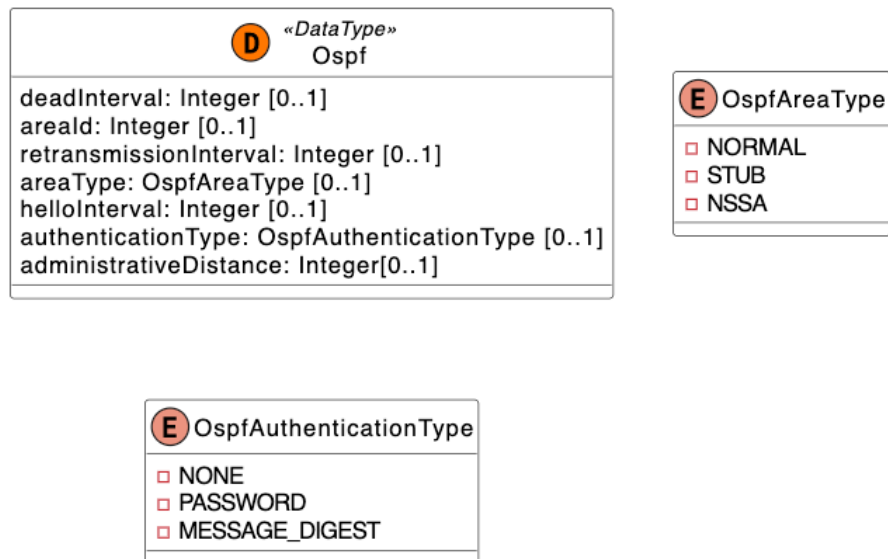


Figure 19-Ospf Model

15.5.1 Ospf

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description

areaid	Integer	0..1	Area ID (0-429967295), normally expressed as an IPv4 address.
areaType	OspfAreaType	0..1	OSPF Area Type enumeration.
authenticationType	OspfAuthenticationType	0..1	OSPF Authentication Type.
helloInterval	Integer	0..1	Hello interval (0-429967295, in seconds).
deadInterval	Integer	0..1	Dead interval (0-429967295, in seconds).
retransmissionInterval	Integer	0..1	Retransmit interval (integer greater than 0, in seconds).
administrativeDistance	Integer	0..1	Administrative distance (integer greater than 0).

Table 26-Ospf Attributes

15.5.2 OspfAreaType

OSPF Area Type enumeration. Reference MEF 61.1 Section 12.7.2 OSPF and MEF 70.2 Section 11.6.3 OSPF.

File: /sdWan/common/ipCommon.yaml

- NORMAL
- STUB
- NSSA

15.5.3 OspfAuthenticationType

OSPF Authentication Type enumeration.

File: /sdWan/common/ipCommon.yaml

- NONE
- PASSWORD
- MESSAGE_DIGEST

15.6 PerformanceTimeIntervals

Specifies a set of time intervals used in the determination of the performance of all Paths in the SWVC. The value of the Service Attribute is a 3-tuple <evalInterval,arrivalInterval,irDuration>. Reference MEF 70.2 Section 9.7 SWVC Performance Time Intervals Service Attribute.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
evalInterval	Integer	1	The interval in milliseconds over which the Performance Metrics specified in the performance Policy Criterion are evaluated.
arrivalIntervals	Integer	1	The difference in arrival times (specified in milliseconds) at the Ingress UNI between two packets used to compute the Mean One-Way Packet Delay Variation.
irDuration	Integer Maximum = 1000	1	The time interval in milliseconds over which the information rate is determined in the evaluation of the BANDWIDTH Policy Criterion. The information rate is determined over any time interval of duration. Should not exceed 1000 milliseconds.

Table 27-PerformanceTimeIntervals Attributes

15.7 RateLimiter

A 3-tuple, <name, commit, limit> where:

- `name` is an Identifier String that is name of Rate Limiter.
- `commit` is the threshold rate (bits per second) at or below which the SD-WAN Service Provider commits to deliver packets in the Bandwidth Flow with high probability under all traffic conditions.
- `limit` is the threshold information rate (bits per second) above which the Service Provider does not deliver IP Packet in the Bandwidth Flow under any conditions.

Reference MEF 70.2 Section 9.9 SWVC List of Rate Limiters Service Attribute.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
name	String	1	Identifier of Rate Limiter.
commit	InformationRate	1	Threshold rate at or below which the SD-WAN Service Provider commits to deliver packets in the Bandwidth Flow with high probability under all traffic conditions.
limit	InformationRate	1	Threshold information rate above which the Service Provider does not deliver IP Packet in the Bandwidth Flow under any conditions.

Table 28-RateLimiter Attributes

15.8 Routing Protocols Model

The UNI Routing Protocol Service Attribute specifies the specifies the routing protocols and associated parameters that are used to exchange IP routes across the UNI. The value is a list of protocols (possibly empty), where each entry consists of the protocol name (one of Static, BGP, or OSPF), the type of routes that will be exchanged (one of IPv4, IPv6 or Both), and a set of additional parameters as specified in the subsections below. According to [R109] in MEF 61.1 and [R139] in MEF 70.2 the value of the UNI Routing Protocols Service Attribute **MUST NOT** contain more than one entry for the same protocol name, except when there are exactly two entries with a given protocol name, one with route type IPv4 and one with route type IPv6. This means per given type of routing protocols one out of four possible sets of configurations can be provided: IPv4, IPv6, and Both. To model that in the API resource model additional types have been added *RoutingProtocolsXXXOptions* that has three mutually exclusive attributes: *ipv4*, *ipv6*, *both*, respectively to handle this requirement.

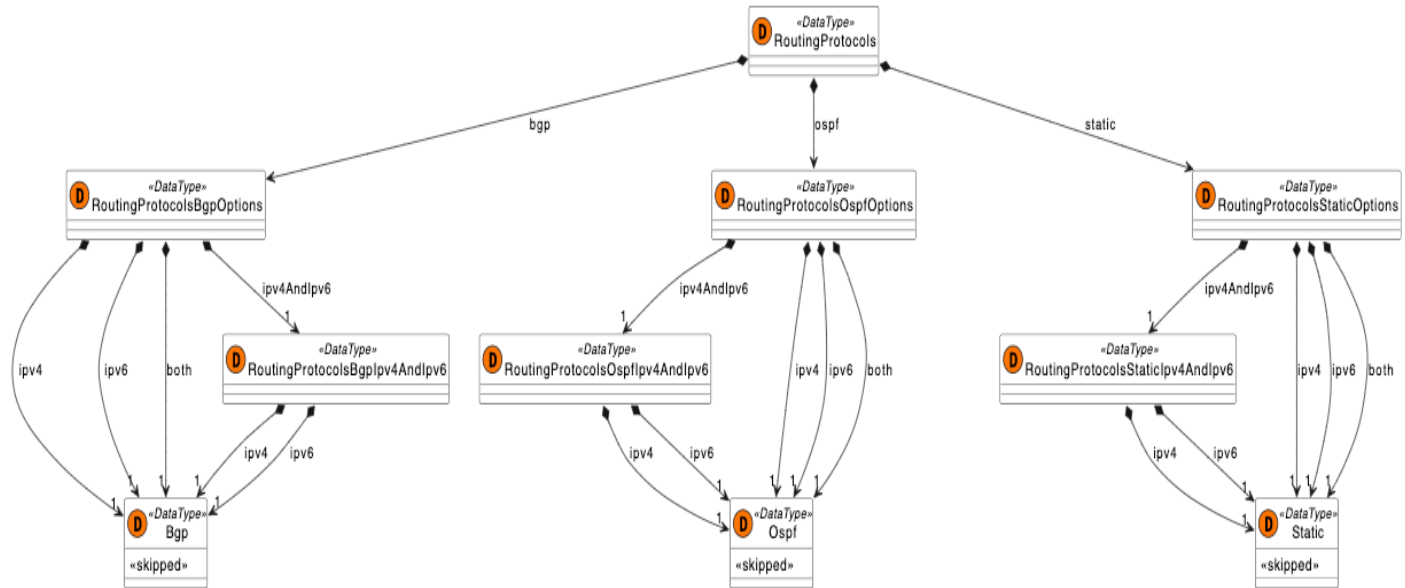


Figure 20-RoutingProtocols Model

15.8.1 RoutingProtocols

Data type to support routing protocols and associated parameters that are used to exchange IP routes across a UNI. The value is a list of protocols (possibly empty), where each entry consists of the protocol name (one of Static, BGP, or OSPF) the type of routes that will be exchanged (one of IPv4 or IPv6 or Both) and a set of additional parameters as specified. Reference MEF 61.1 Section 12.7 UNI Routing Protocols Service Attribute or MEF 70.2 Section 11.6 SD-WAN UNI Routing Protocols Service Attribute.

File: /common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
static	RoutingProtocolStaticOptions	0..1	Static routing configuration options.
bgp	RoutingProtocolBgpOptions	0..1	BGP routing protocol configuration options.
ospf	RoutingProtocolOspfOptions	0..1	OSPF routing protocol configuration options.

Table 29-RoutingProtocol Attributes

15.8.2 RoutingProtocolsBgpIpv4AndIpv6

Configuration for exchanging IPv4 and IPv6 types of routes.

File: /common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4	Bgp	1	Configuration for exchanging IPv4 types of routes.
ipv6	Bgp	1	Configuration for exchanging IPv6 types of routes.

Table 30-RoutingProtocolsBgpIpv4AndIpv6 Attributes

15.8.3 RoutingProtocolsBgpOptions

BGP routing protocol configuration options. The configuration of the BGP can be provided for the following type of routes that will be exchanged:

- ipv4, or
- ipv6, or
- both, or
- ipv4 and ipv6

File: /common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4	Bgp	0..1	Configuration for exchanging IPv4 types of routes.
ipv6	Bgp	0..1	Configuration for exchanging IPv6 types of routes.
ipv4Andipv6	RoutingProtocolsBgpIpv4AndIpv6	0..1	Configuration for exchanging IPv4 and IPv6 types of routes.
both	Bgp	0..1	Configuration for exchanging both IPv4 and IPv6 types of routes.

Table 31-RoutingProtocolsBgpOptions Attributes

oneOf:

- required: [ipv4]
- required: [ipv6]
- required: [both]

- required: [ipv4AndIpv6]

15.8.4 RoutingProtocolsOspfIpv4AndIpv6

Configuration for exchanging IPv4 and IPv6 types of routes.

File: /common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4	Ospf	1	Configuration for exchanging IPv4 types of routes.
ipv6	Ospf	1	Configuration for exchanging IPv6 types of routes.

Table 32-RoutingProtocolsOspfIpv4AndIpv6 Attributes

15.8.5 RoutingProtocolsOspfOptions

OSPF routing protocol configuration options. The configuration of the OSPF can be provided for the following types of routes that will be exchanged:

- ipv4, or
- ipv6, or
- both, or
- ipv4 and ipv6

File: /common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4	Bgp	0..1	Configuration for exchanging IPv4 types of routes.
ipv6	Bgp	0..1	Configuration for exchanging IPv6 types of routes.
ipv4Andipv6	RoutingProtocolsOspfIpv4AndIpv6	0..1	Configuration for exchanging IPv4 and IPv6 types of routes.
both	Ospf	0..1	Configuration for exchanging both IPv4 and IPv6 types of routes.

Table 33-RoutingProtocolsOspfOptions Attributes

oneOf:

- required: [ipv4]
- required: [ipv6]
- required: [both]
- required: [ipv4AndIpv6]

15.8.6 RoutingProtocolsStaticIpv4AndIpv6

Configuration for exchanging IPv4 and IPv6 types of routes.

File: /common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4	Static	1	Configuration for exchanging IPv4 types of routes.
ipv6	Static	1	Configuration for exchanging IPv6 types of routes.

Table 34-RoutingProtocolsStaticIpv4AndIpv6 Attributes

15.8.7 RoutingProtocolsStaticOptions

Static routing protocol configuration options. The configuration of the Static can be provided for the following types of routes that will be exchanged:

- ipv4, or
- ipv6, or
- both, or
- ipv4 and ipv6

File: sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ipv4	Static	0..1	Configuration for exchanging IPv4 types of routes.
ipv6	Static	0..1	Configuration for exchanging IPv6 types of routes.
ipv4Andipv6	RoutingProtocolsStaticIpv4AndIpv6	0..1	Configuration for exchanging IPv4 and IPv6 types of routes.
both	Static	0..1	Configuration for exchanging both IPv4 and IPv6 types of routes.

Table 35-RoutingProtocolsStaticOptions Attributes

oneOf:

- required: [ipv4]
- required: [ipv6]
- required: [both]
- required: [ipv4AndIpv6]

15.8.8 ServiceNameAndTransportProtocols

Service names and port numbers are used to distinguish between different services that run over transport protocols such as TCP, UDP, DCCP, and SCTP.

File: sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
serviceName	string	1	Service name.
portNumber	Integer Min: 0 Max: 65535	1	Port number.
transportProtocol	String Enum: • TCP • UDP	1	Transport protocol.

Table 36-ServiceNameAndTransportProtocols Attributes

15.8.9 ServiceNameAndTransportProtocolsOrAny

Data type which allows an ServiceNameAndTransportProtocols or Any.

File: /sdWan/common/ipCommon.yaml			
oneOf:			
required: ServiceNameAndTransportProtocols			
required: Any			
Attribute Name	Type	Multiplicity	Description
subscriberIpv4Address	Ipv4Address	0..1	Subscriber IPv4 Address.

Table 37-ServiceNameAndTransportProtocolsAndAny Attributes

15.9 ServicePerformanceObjectivesReportingPeriods

Service Performance Objectives are evaluated over a series of consecutive time periods. The value of this Service Attribute is a 2-tuple <s,T> where:

- s (startTime) is a time that represents the date and time that the evaluation of Service Performance Objectives starts for the SWVC and its components.
- T (duration) is a time duration.

Reference MEF 70.2 Section 9.8 SWVC Service Performance Objectives Reporting Periods Service Attribute.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
startTime	TimeAndDate	1	Time that represents the date and time that the evaluation of Service Performance Objectives starts for the SWVC and its components.
duration	TimeDuration	1	Time duration.

Table 38-ServicePerformanceObjectivesReportingPeriods Attributes

15.10 Static Model

When an entry in the (SD-WAN) UNI Routing Protocols list is for Static, the IP Prefixes used in the Subscriber Network that are reachable via this UNI are specified as additional parameters in the entry. These are known as Static IP Prefixes. Reference MEF 61.1 Section 12.7.1 Static and MEF 70.2 Section 11.6.1.

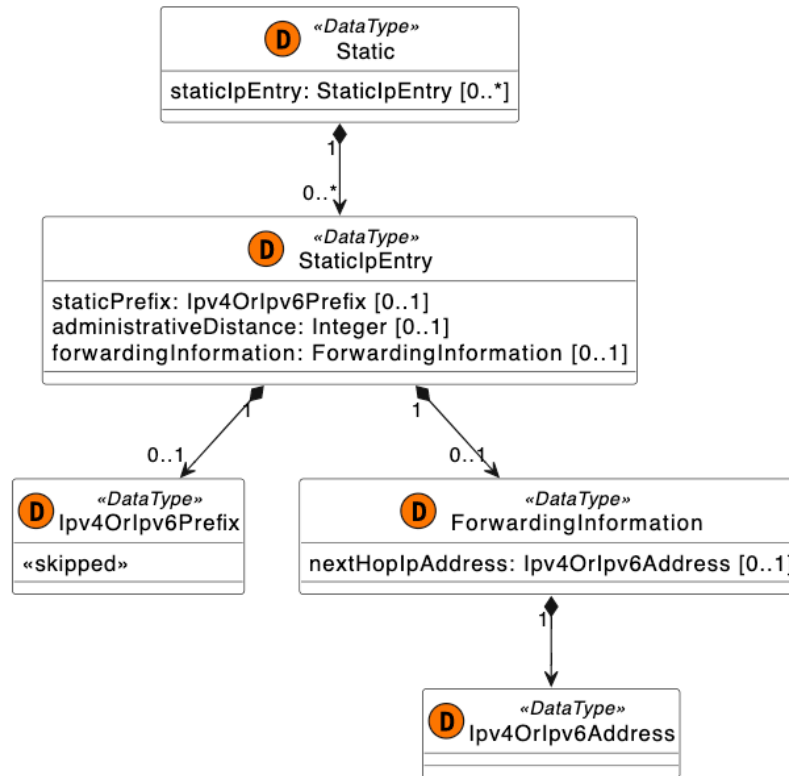


Figure 21-Static Model

15.10.1 Static

Editor Note 2: Normative

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
staticIpEntry	StaticIpEntry	1..*	Static IP address entry.

Table 39-Static Attributes

15.10.2 StaticIpEntry

StaticIpEntry data type including IPv4/IPv6 prefixes, forwarding information and administrative distance.

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
staticIpEntry	StaticIpEntry	1..*	Static IP address entry.

Table 40-Static Attributes

15.11 SubscriberAndSpLoopbackAddresses

A list of pairs of IP addresses, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses.

File: /sdWan/common/ipCommon.yaml			
Attribute Name	Type	Multiplicity	Description
subscriberLoopbackAddress	Ipv4OrIpv6Address	0..1	Subscriber's loopback address for BGP establishing a session.
spLoopbackAddress	Ipv4OrIpv6Address	0..1	Service Provider's loopback address for BGP establishing a session.

Table 41-SubscriberAndSpLoopbackAddresses Attributes

15.12 SwVcIdentifier

File: /sdWan/sdWanCommon.yaml

The value of SWVC Identifier Service Attribute is a string that is used by the Subscriber and the Service Provider to uniquely identify an SWVC. Reference MEF 70.2 Section 9.1 SWVC Identifier Service Attribute.

- type: string
- maxLength: 53
- pattern: "[/x20-\x7F]+"

15.13 SwVcEndPointIdentifier

File: /sdWan/sdWanCommon.yaml

SWVC End Point Identifier as described in MEF 70.2 Section 10.1 SWVC End Point Identifier Service Attribute. A string that is used to allow the Subscriber and Service Provider to uniquely identify the association of the SWVC with a UNI.

- type: string
- maxLength: 53

- pattern: “[/x20-\x7F]+”

15.14 TimeAndDate

Date type for Time and Date in UTC.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
day	Integer	1	This denotes the day.
hour	Integer	1	This denotes the hour.
minute	Integer	1	This denotes the minute.
month	Integer	1	This denotes the month.
second	Integer	1	This denotes the second.
year	Integer	1	This denotes the year.

Table 42-TimeAndDate Attributes

15.15 TimeDuration

This data type is used to describe durations expressed as a 2-tuple, (value, units). The units from nanoseconds to years.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
timeDurationValue	integer	1	The value of the duration. For example, if the duration is 20 ms, this element is 20.
timeDurationUnits	TimeDurationUnits	1	The unit of measure in duration. For example, if an interval is 2 ms, this element is NS.

Table 43-TimeDuration Attributes

15.16 TimeDurationUnits

File: /sdWan/sdWanCommon.yaml

1071 The unit of measure in the duration. For example, if an interval is 2ms, this element is MS.

- 1072 • NS:
 - 1073 ○ Nanoseconds
- 1074 • US:
 - 1075 ○ Microseconds
- 1076 • MS
 - 1077 ○ Milliseconds
- 1078 • SEC
 - 1079 ○ Seconds
- 1080 • MIN
 - 1081 ○ Minutes
- 1082 • HOUR
- 1083 • DAY
- 1084 • WEEK
- 1085 • MONTH
- 1086 • YEAR
- 1087

16 IP Routing Protocol Common Resources and Types

This section focuses on IP routing protocol common resources and types used by the Service Attributes. All of these are structured to support a variety of IP resources used by the SD-WAN Service model. This section details the data types and enumerations that are used by the SD-WAN Service model.

16.1 AddressFamilyIpv4Ipv6

Schema File Name: *schema/common/ipRoutingProtocolsCommon.yaml*

Specifies whether the session is established over IPv4 or IPv6.

- IPV4
- IPV6

16.2 Bgp

When an entry in the (SD-WAN) UNI Routing Protocol is for BGP, BGP as specified in RFC 4271 is used across the UNI to exchange routing information. Reference MEF 61.1 Section 12.7.3 BGP and MEF 70.2 Section 11.6.2 BGP. Figure 22 depicts the BGP routing protocol configuration model.

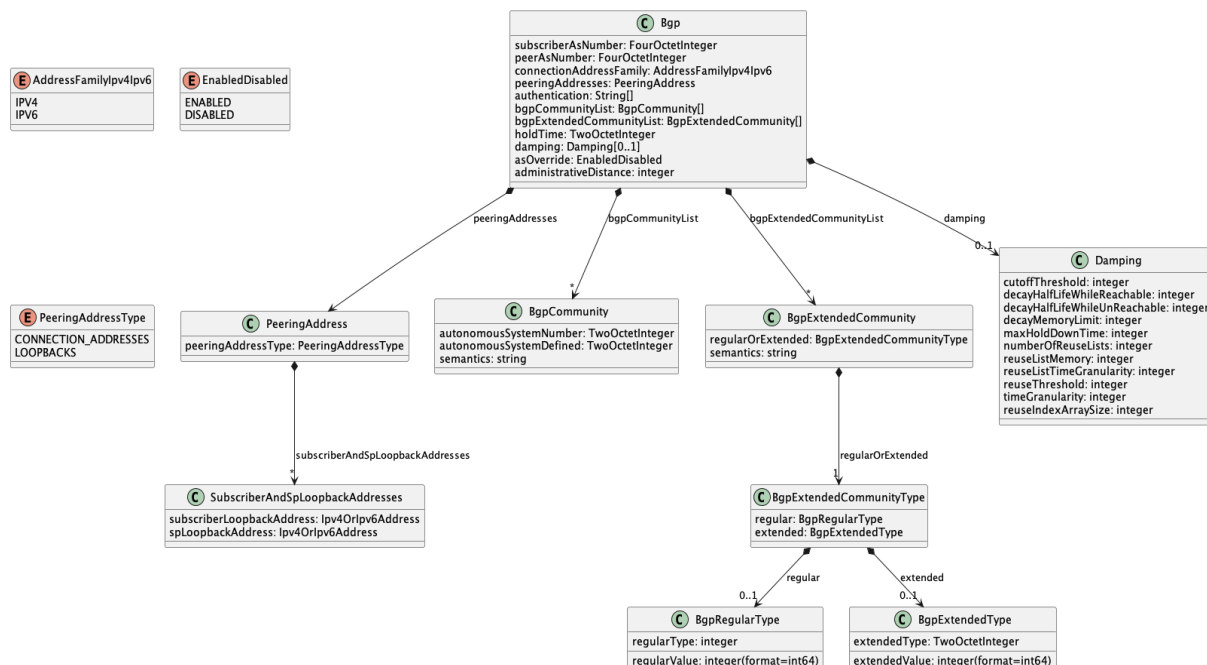


Figure 22-Bgp Model

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml			
Attribute Name	Type	Multiplicity	Description
subscriberAsNumer	FourOctetInteger	1	BGP Subscriber Autonomous System number.
peerAsNumber	FourOctetInteger	1	BGP Peer Autonomous System Number.
connectionAddressFamily	AddressFamilyIpv4I pv6	1	Connection Address Family (IPv4 or IPv6).
peeringAddresses	PeeringAddresses	1	Peering Addresses.
authentication	String	1..*	BGP Authentication (None or MD5 plus a password).
bgpCommunityList	BgpCommunity	0..*	Used to control which routers are accepted, preferred, distributed, or advertised.
bgpExtendedCommunityList	BgpExtendedComm unity	0..*	Mechanism for labeling information carried in BGP-4. Provide enhancement over existing BGP Community Attribute: An extended range, the addition of type field.
holdTime	TwoOctetInteger	1	Hold time in seconds. Indicates the agreed Hold Time used for BGP sessions. The possible values are 0 or an integer in the range 3 -65535.
damping	Damping	0..1	Route flap damping. When the Damping parameter is NONE, the attribute is NOT set. When not NONE a single set of parameters described in Section 4.3 of RFC 2430 MUST be agreed.
asOverride	EnabledDisabled	1	Autonomous System Override. The SP (or Operator) can overwrite instances of the Subscriber's AS Number in the AS Path with their own AS Number, when advertising routes to the Subscriber. This needs to be explicitly agreed between the SP and the Subscriber, and/or between an SP/SO and an Operator.
administrativeDistance	Integer	1	BGP Administrative Distance.

1104

Table 44-Bgp Attributes

1105 16.3 BgpCommunity

1106 A community is a group of destinations which share some common property. Each autonomous
1107 system administrator may define which communities a destination belongs to.

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml			
Attribute Name	Type	Multiplicity	Description
autonomousSystemNumber	TwoOctetInteger	1	The first two octets encoding the Autonomous System value.
autonomousSystemDefined	TwoOctetInteger	1	The remaining octets.
semantics	String	1	Text describing how the Seller will handle routes tagged with this Community.

1108

Table 45-BgpCommunity Attributes

1109 16.4 BgpExtendedCommunity

1110 This attribute provides a mechanism for labeling information carried in BGP-4. These labels can
1111 be used to control the distribution of this information, or for other applications.

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml

Required:

- regularOrExtended
- semantic

Attribute Name	Type	Multiplicity	Description
regularOrExtended	BgpExtendedCommunityType	1	Indicates whether the BGP Extended Community is of Regular Type or Extended Type.
semantics	String		Text describing how the Seller will handle routes tagged with this Community.

Table 46-BgpExtendedCommunity Attributes

1113 16.5 BgpExtendedCommunityType

1114 Options for BGP Extended Community types. Either Regular Type or Extended Type.

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml

Attribute Name	Type	Multiplicity	Description
regular	BgpRegularType	1	BGP Regular Type.
extended	BgpExtendedType	1	BGP Extended Type.
minProperties: 1 maxProperties: 1			

Table 47-BgpExtendedCommunityType Attributes

1116 16.6 BgpRegularType

1117 This resource provides a mechanism for labeling the information carrier in.

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml

Attribute Name	Type	Multiplicity	Description
regularType	Integer	1	Regular Type Field, 1 octet length. • Minimum: 0 • Maximum: 255
regularValue	Integer Format: int64 Minimum: 0 Maximum: 7205759403792794 0	1	Octets 2-8 of the value part of the address. Used in case only Regular Type is provided.

Table 48-BgpRegularType Attributes

1119 16.7 BgpExtendedType

1120 This resource provides a mechanism for labeling the information carrier in.

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml			
Attribute Name	Type	Multiplicity	Description
extendedType	TwoOctetInteger	1	Extended Type Field 2, 2 octets length.
extendedValue	Integer Format: int64 Minimum: 0 Maximum: 281474976710655	1	Octets 3-8 of the value part of the address. Used in case only Extended Type is provided.

1121 **Table 49-BgpExtendedType Attributes**

1122 16.8 Damping

1123 Damping parameters as defined in RFC 2439 [4] BGP Route Flap Damping, Section 4.2

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml			
Attribute Name	Type	Multiplicity	Description
cutoffThreshold	Integer	1	This value is expressed as a number of route withdrawals. It is the value above which a route advertisement will be suppressed.
decayHalfLifeWhileReachable	Integer	1	This value is the time duration in seconds during which the accumulated stability figure of merit will be reduced by half if the route is considered reachable (whether suppressed or not).
decayHalfLifeWhileUnreachable	Integer	1	This value is the time duration in seconds during which the accumulated stability figure of merit will be reduced by half if the route is considered unreachable. If not specified or set to zero, no decay will occur while a route remains unreachable.
decayMemoryLimit	Integer	1	This is the maximum time (in seconds) that any memory of previous instability will be retained given that the route's state remains unchanged, whether reachable or unreachable. This parameter is generally used to determine array sizes.
maxHoldDownTime	Integer	1	This value is the maximum time a route can be suppressed no matter how unstable it has been prior to this period of stability. In seconds.

numberOfReuseLists	Integer	1	This is the number of reuse lists. It may be determined from reuse-list-max or set explicitly.
reuseListMemoryReuseListMax	Integer	1	This is the time (in seconds) value corresponding to the last reuse list. This may be the maximum value of T-hold for all parameter sets of may be configured.
reuseListTimeGranularity	Integer	1	This is the time (in seconds) interval between evaluations of the reuse lists. Each reuse lists corresponds to an additional time increment.
reuseThreshold	Integer	1	This value is expressed as a number of route withdrawals. It is the value below which a suppressed route will now be used again.
timeGranularity	Integer	1	This is the time granularity in seconds used to perform all decay computations.
reuseIndexArraySize	Integer	1	This is the size of reuse index arrays. This size determines the accuracy with which suppressed routes can be placed within the set of reuse lists when suppressed for a long time.

1124 **Table 50-Damping Attributes**

1125 **16.9 ForwardingInformation**

1126 Forwarding information, consisting of either a nexthop IP address in the Subscriber Network (if
1127 the access medium is multipoint capable, e.g., Ethernet), or a specific UNI Access Link (if the
1128 access medium is strictly point-to-point, e.g., HDLC, PPP over DSL). *NOTE: UNI Access Link is*
1129 *applicable only to IP services.*

Schema File Name: schema/common/ip/ipRoutingProtocolsCommon.yaml			
Attribute Name	Type	Multiplicity	Description
nextHopIpAddress	Ipv4OrIpv6Address	0..1	Next hop IP address.

1130 **Table 51-ForwardingInformation Attributes**

17 SD-WAN Common Resources and Types

This section is structure like the previous section but focuses on common resources and types used by the Service Attributes. Most of these are structured to support a variety of SD-WAN Services. This section details the data types and enumerations that are used by the SD-WAN Service model.

17.1 ApplicationFlowBusinessImportanceLevels

Specifies an ordered list of labels that can be assigned to Application Flows to indicate importance of each Application Flow. Reference MEF 70.2 Section 9.11 SWVC List of Application Flow Business Importance Levels Service Attribute.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
implist	ApplicationFlowBusinessImportanceLevelIdentifier	1..*	A non-empty list of Identifier Strings that are the business importance labels ordered from highest importance to lowest.
afbilDefault	ApplicationFlowBusinessImportanceLevelIdentifier	1	Default. Is one of the labels in implist that represents the business importance of Application Flows that do not have a business importance assigned by Policy.

Table 52-ApplicationFlowBusinessImportanceLevels Attributes

17.2 ApplicationFlowBusinessImportanceLevelIdentifier

Schema File Name: /sdWan/sdWanCommon.yaml

The Identifier string of business importance labels.

- type: string
- maxLength: 53
- pattern: "[/x20-\x7F]+"

17.3 ApplicationFlowSpecification

Specifies the Application Flows that can be recognized by the SD-WAN service and information about how to identify IP Packets in each Application Flow. A non-empty ordered list of 3-tuples <AFname, AFCritList, AFGGroup>.

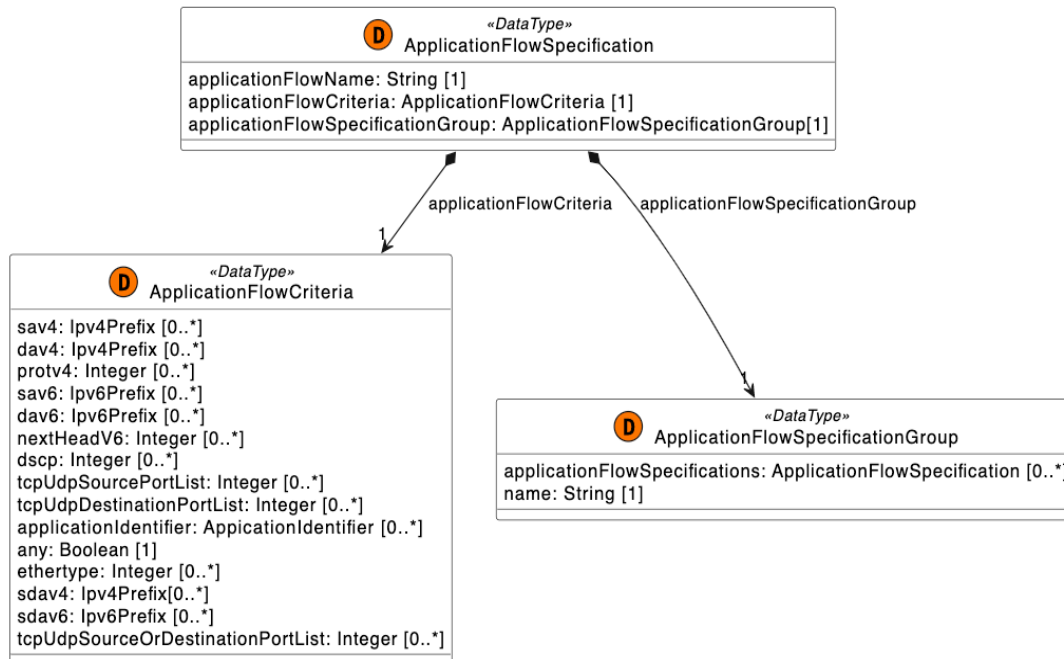


Figure 23-Application Flows and Application Flow Groups Model

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
applicationFlowName	String	1	Identifier String that is used to refer to Application Flow Specification. Reference MEF 70.2 Section 9.14 SWVC List of Application Flow Specifications Service Attribute.
applicationFlowCriteria	ApplicationFlowCriteria	1..*	Pointer to associated ApplicationFlowCriteria. Reference MEF 70.2 Section 9.14 SWVC List of Application Flow Specifications Service Attribute.

applicationFlowSpecificationGroup	ApplicationFlowSpecificationGroup	1	Pointer to associated ApplicationFlowSpecificationGroup. Reference MEF 70.2 Section 9.14 SWVC List of Application Flow Specifications Service Attribute.
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Table 53-ApplicationFlowAttributes

17.4 ApplicationFlowCriteria

The ApplicationFlowCriteria describes the standard criteria that can be used to describe an Application Flow, as defined in MEF 70.2 Table 7 – Application Flow Criteria – Support Required and Table 9 Application Flow Criteria – Support Recommended.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
sav4	Ipv4Prefix	0..*	IPv4 Source Address. List of IPv4 Prefixes. Reference MEF 70.2 Section 9.14 Table-7.
dav4	Ipv4Prefix	0..*	IPv4 Destination Address. List of IPv4 Prefixes. Reference MEF 70.2 Section 9.14 Table-7.
protv4	ProtocolV4	0..*	IPv4 Protocol List. List of integers in the range 0 to 255 or a list of keywords for IANA Protocol Numbers Registry. Reference MEF 70.2 Section 9.14 Table-7.
sav6	Ipv6Prefix	0..*	IPv6 Source Address. List of IPv6 Prefixes. Reference MEF 70.2 Section 9.14 Table-7.
dav6	Ipv6Prefix	0..*	IPv6 Destination Address. List of IPv6 Prefixes. Reference

			MEF 70.2 Section 9.14 Table-7.
nextHeadV6	NextHeadV6	0..*	IPv6 Next Header List. Reference MEF 70.2 Section 9.14 Table-7.
dscp	Dscp	0..*	List of Integers in the range 0 to 63. Reference MEF 70.2 Section 9.14 Table-7.
sport	PositiveInteger	0..*	Transport Source Port. List of integers in the range 0 to 65535 or a list of service names from IANA Service Name and Port Number Registry. Reference MEF 70.2 Section 9.14 Table-7/8.
dport	PositiveInteger	0..*	List of integers in the range 0 to 65535 or a list of service names from IANA Service Name and Port Number Registry. Reference MEF 70.2 Section 9.14 Table-7.
appld	ApplicationIdentifier	0..*	Custom match including heuristic/algorithmic matching. MEF 70.2 Section 9.14 Table-7.
any	Boolean	1	Match any IP Packet. Reference MEF 70.2 Section 9.14 Table-7.
ethertype	Ethertype	0..1	Ethertype. Integer in the range 0x600 to 0xffff. Reference MEF 70.2 Section 9.14 Table-8.
sdav4	Ipv4Prefix	0..*	Match a list of values for either the source or designation

			address. Reference MEF 70.2 Section 9.14 Table-8.
sdav6	Ipv6Prefix	0..*	Same as previous.
sdport	PositiveInteger	0..*	Match a list of values for either the source or destination port.

Table 54-ApplicationFlowCriteria Attributes

17.5 ApplicationFlowSpecificationGroup

Provides a mechanism for associating a Policy with multiple Application Flows. Reference MEF 70.2 Section 9.13 SWVC List of Application Flow Specification Groups Service Attribute.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
applicationFlowSpecifi cations	ApplicationFlowSpe cification	0..*	Associated Application Flows.
name	String	1	Application Flow Group name.

Table 55-ApplicationFlowGroup Attributes

17.6 ApplicationIdentifier

A method for referring to named packet matching definitions (both simple and complex) defined by a Service Provider.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
applicationIdentifier	String	1	The Application Identifier.
additionalArguments	String	0..*	Additional arguments required (beyond the Identifier).
applicationMatch	String	0..*	Details of the applications and application traffic matched by the APPID.

Table 56-ApplicationIdentifier Attributes

NOTE: The ApplicationIdentifier could have additional complex type definition but is currently defined using Strings.

17.7 BandwidthPolicyCriterion

Provides a method to express the intended bandwidth requirements for an Application Flow, and the probability of packet delay or discard in the face of varying bandwidth contention for Underlay Connectivity Service resources. Reference MEF 70.2 Section 9.12.2.10 BANDWIDTH Ingress Policy Criterion.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
unnamedRateLimiter	UnnamedRateLimiter	0..1	An unnamed Rate Limiter.
rateLimiter	RateLimiter	0..1	A named Rate Limiter specified in the SWVC List of Rate Limiters.

Table 57-BandwidthPolicyCriterion Attributes

one of:

- required: [unnamedRateLimiter]
- required: [rateLimiter]

17.8 CipherSuite

A Cipher suite includes the protocol name (i.e., TLS), the Key Exchange algorithm (e.g., DHE_RSA), with the encryption algorithm (e.g., AES_128_GCM) and the Message Authentication algorithm (e.g., SHA256). Reference IANA, TLSCipher Suites Registry, <https://www.iana.org/assignments/tls-parameters/tls-parameters.xhtml>.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
value	String	1	Cipher value.
name	String	1	Enumeration (in future) representing Cipher Suite Name.

Table 58-CipherSuite Attributes

17.9 Dnf

Domain Name Filtering is defined as the Security Function that determines whether an Application Flow, or subset of an Application Flow, contains domain names. This data type represents a Domain Name for filtering.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
domainName	String	1	Domain name.

Table 59-Dnf Attributes

17.10 DnfOrNone

Data type to allow the selection of Dnf or *None*. Note that the value *None* indicates that the IPPF Security Function is *Disabled* for all Application Flows to which this Security Policy is applied. Reference [MEF 88 Section 6.5 DNS Protocol Filtering Parameter](#).

Schema File Name: /sdWan/sdCommon.yaml			
one of:			
-required: [dnfNone]			
-required: [dnfSelect]			
Attribute Name	Type	Multiplicity	Description
dnfNone	String Enum: • NONE	1	Select NONE for no DNF.
dnfSelect	DomainNameFilter	1	Select DNF.

Table 60-DnfOrNone Attributes

17.11 Dpf

Service Provider maintained list of 5-tuple for IP filtering. Reference [MEF 88 Section 9.2 DNS Protocol Filtering](#).

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
sav4	Ipv4AddressOrAny	1..*	A list of IPv4 source addresses or a value of ANY meaning any IPv4 address.

dav4	Ipv4AddressOrAny	1..*	A list of IPv4 destination addresses or a value of ANY meaning any IPv4 address.
sav6	Ipv6AddressOrAny	1..*	A list of IPv6 source addresses or a value of ANY meaning any IPv6 address.
dav6	Ipv6AddressOrAny	1..*	A list of IPv6 destination addresses or a value of ANY meaning any IPv6 address.
dnsMessageType	String	1	One of the DNS messages, as specified in RFC 6895 [9].

Table 61-Dpf Attributes

17.12 DpfOrNone

Data type to allow the selection of Ippf or *None*. Note that the value *None* indicates that the IPPF Security Function is *Disabled* for all Application Flows to which this Security Policy is applied. Reference [MEF 88 Section 6.5 DNS Protocol Filtering Parameter](#).

Schema File Name: /sdWan/sdCommon.yaml one of: -required: [dpfNone] -required: [dpfSelect]			
Attribute Name	Type	Multiplicity	Description
dpfNone	String Enum: • NONE	1	Select NONE for no DPF.
dpfSelect	DnsProtocolFilter	1	Select DPF.

Table 62-DpfOrNone Attributes

17.13 DnsProtocolFilter

Provides for the filtering of Application Flows with respect to IP addresses of DNS servers and DNS message types. Reference [MEF 88 Section 6.5 DNS Protocol Filtering Parameter](#).

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description

dpfBlockList	Dpf	1..*	List of match criteria in DPF Block List or None.
dpfAllowList	Dpf	1..*	List of match criteria in DPF Allow List or None.
dpfQuarantineList	Dpf	1..*	List of match criteria in DPF Quarantine List or None.
nm	String	1	Parameter that determines the behavior when a subset of the Application Flow does not match criteria entry on any of the DPF List. The possible values are ALLOW or BLOCK.
duration	TimeDuration	1	Duration of the time between updates of the IPPF security threat database.

Table 63-DnsProtocolFilter Attributes

17.14 DomainNameFilter

Provides for the filtering of Application Flows with respect to domain names. [Reference MEF 88 Section 6.6 Domain Name Filtering Parameter.](#)

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
dnfBlockList	Dnf	1..*	List of match criteria in DNF Block List or None.
dnfAllowList	DnfOrNone	1..*	List of match criteria in DPF Allow List or None.
dnfQuarantineList	DnfOrNone	1..*	List of match criteria in DPF Quarantine List or None.
nm	String	1	Parameter that determines the behavior when a subset of the Application Flow does not match criteria entry on any of the DPF List. The possible values are ALLOW or BLOCK.
duration	TimeDuration	1	Duration of the time between updates of the IPPF security threat database.

Table 64-DomanProtocolFilter Attributes

17.15 IanaProtocols

Protocol Numbers values. Reference IANA Protocol Numbers.

Schema File Name: /sdWan/sdWanCommon.yaml

- HOPOPT
- ICMP
- IGMP
- GGP
- IPV4
- ST
- TCP
- CBT
- EGP
- IGP
- BBN_RCC_MON
- NVP_II
- PUP
- ARGUS
- EMCON
- XNET
- CHAOS
- UDP
- MUX
- DCN_MEAS
- HMP
- PRM
- XNS_IDP
- TRUNK_1
- TRUNK_2
- LEAF_1
- LEAF_2
- RDP

- 1243 • IRTP
- 1244 • ISO_TP4
- 1245 • NETBLT
- 1246 • MFE_NSP
- 1247 • DCCP
- 1248 • 3PC
- 1249 • IDPR
- 1250 • XTP
- 1251 • DDP
- 1252 • IDRP_CMTP
- 1253 • TP_PLUS_PLUS
- 1254 • IL
- 1255 • IPV6
- 1256 • SDRP
- 1257 • IPV6_ROUTE
- 1258 • IPV6_FRAG
- 1259 • IDRP
- 1260 • RSVP
- 1261 • GRE
- 1262 • DSR
- 1263 • BNA
- 1264 • ESP
- 1265 • AH
- 1266 • I_NLSP
- 1267 • SWIPE
- 1268 • NARP
- 1269 • MOBILE
- 1270 • TLSP
- 1271 • SKIP
- 1272 • IPV6_ICMP
- 1273 • IPV6_NO_NXT
- 1274 • IPV6_OPTS
- 1275 • CFTP
- 1276 • SAT_EXPAK
- 1277 • KRYPTOLAN

- 1278 • RVD
- 1279 • IPPC
- 1280 • SAT_MON
- 1281 • VISA
- 1282 • IPCV
- 1283 • CPNX
- 1284 • CPHB
- 1285 • WSN
- 1286 • PVP
- 1287 • BR_SAT_MON
- 1288 • SUN_ND
- 1289 • WB_MON
- 1290 • WB_EXPAK
- 1291 • ISO_IP
- 1292 • VMTP
- 1293 • SECURE_VMTP
- 1294 • VINES
- 1295 • IPTM
- 1296 • NSFNET-IGP
- 1297 • DGP
- 1298 • TCF
- 1299 • EIGRP
- 1300 • OSPFIGP
- 1301 • SPRITE_RPC
- 1302 • LARP
- 1303 • MTP
- 1304 • AX_25
- 1305 • IPIP
- 1306 • MICP
- 1307 • SCC_SP
- 1308 • ETHERIP
- 1309 • ENCAP
- 1310 • GMTP
- 1311 • IFMP
- 1312 • PNNI

- 1313 • PIM
- 1314 • ARIS
- 1315 • SCPS
- 1316 • QNX
- 1317 • A_N
- 1318 • IP_COMP
- 1319 • SNP
- 1320 • COMPAQ_PEER
- 1321 • IPX_IN_IP
- 1322 • VRRP
- 1323 • PGM
- 1324 • L2TP
- 1325 • DDX
- 1326 • IATP
- 1327 • STP
- 1328 • SRP
- 1329 • UTI
- 1330 • SMP
- 1331 • SM
- 1332 • PTP
- 1333 • ISIS_OVER_IPV4
- 1334 • FIRE
- 1335 • CRTP
- 1336 • CRUDP
- 1337 • SSCOPMCE
- 1338 • IPLT
- 1339 • SPS
- 1340 • PIPE
- 1341 • SCTP
- 1342 • FC
- 1343 • RSVP_E2E_IGNORE
- 1344 • MOBILITY_HEADER
- 1345 • UDP_LITE
- 1346 • MPLS_IN_IP
- 1347 • MANET

- 1348 • HIP
- 1349 • SHIM6
- 1350 • WESP
- 1351 • ROHC
- 1352 • ETHERNET
- 1353 • AGGFRAG
- 1354 • UNASSIGNED
- 1355 • EXPERIMENTATION
- 1356 • RESERVED

1357 17.16 IanaProtocolsIntegers

1358 Protocol number integers.

1359 *Schema File Name: /sdWan/sdWanCommon.yaml*

1360

1361 type: integer

1362 minimum: 0

1363 maximum: 255

1364 17.17 IanaProtocolsIntegerOrKeyword

1365 Data type which allows Iana Protocols Integer values or keywords.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
ianaProtocolKeyword	IanaProtocols	1	Protocol Numbers values. Reference IANA, Protocol Numbers.
ianaProtocolInteger	IanaProtocolsIntegers	1	Protocol Number as Integer.

1366 **Table 65-IanaProtocolsIntegerOrKeyword Attributes**

1367 17.18 Ippf

1368 Service Provider maintained list of 8-tuples for IP filtering. Reference MEF 88 Section 9.1 IP, Port
1369 and Protocol Filtering.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description

sav4	Ipv4AddressOrAny	1..*	A list of IPv4 source addresses or a value of ANY meaning any IPv4 address.
dav4	Ipv4AddressOrAny	1..*	A list of IPv4 destination addresses or a value of ANY meaning any IPv4 address.
protov4	IanaProtocolsIntegerOrKeyword	1..*	The IPv4 protocol list, is a list of integers in range 0 to 255 or a list of keywords from IANA Protocol Numbers, or a mix of integers and keywords.
sav6	Ipv6AddressOrAny	1..*	A list of IPv6 source addresses or a value of ANY meaning any IPv6 address.
dav6	Ipv6AddressOrAny	1..*	A list of IPv6 destination addresses or a value of ANY meaning any IPv6 address.
nextheadv6	IanaProtocolsIntegerOrKeyword	1..*	The IPv6 protocol list, is a list of integers in the range 0 to 255 or a list of keywords from IANA, Protocol Numbers or a mix of integers and keywords.
sport	ServiceNameAndTransportProtocolsOrAny	1..*	Transport Source Port is a list of integers in the range 0 to 65535 or a list of keywords from IANA, Protocol Numbers, or a mix of integers and keywords. Any could be used to indicate that any source port number is on the list.
dport	ServiceNameAndTransportProtocolsOrAny	1..*	Transport Destination Port is a list of integers in the range 0 to 65535 or a list of keywords from IANA, Protocol Numbers, or a mix of integers and keywords. Any could be used to indicate that any source port number is on the list.

Table 66-Ippf Attributes

1370

17.19 IppfOrNone

Data type to allow the selection of Ippf or *None*. Note that the value *None* indicates that the IPPF Security Function is *Disabled* for all Application Flows to which this Security Policy is applied. Reference MEF 88 Section 6.4 IP, Port and Protocol Filtering Parameter.

Editor Note 3: Change to MEF 138

Schema File Name: /sdWan/sdCommon.yaml one of: -required: [ippfNone] -required: [ippfSelect]			
Attribute Name	Type	Multiplicity	Description
ippfNone	String Enum: NONE	1	Select NONE for no IPPF.
ippfSelect	IpPortProtocolFilter	1	Select IPPF.

Table 67-IppfOrNone Attributes

17.20 IpPortProtocolFilteringSecurityFunctionPolicy

Provides for the filtering of Application Flows with respect to IP addresses, port numbers and IP protocols. Reference MEF 138, Section 9.4 IP, Port and Protocol Filtering (IPPF) Security Function Policy.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
policyId		1	The identifier for a given IPPF Security Function Policy.
ippfBlockList	Ippf	1..*	List of match criteria in IPPF Block List or None.
ippfAllowList	Ippf	1..*	List of match criteria in IPPF Allow List or None.
ippfQuarantineList	Ippf	1..*	List of match criteria in IPPF Quarantine List or None.
nm	String	1	Parameter that determines the behavior when a subset of the Application Flow does not match criteria entry on any of the IPPF

			List. The possible values are ALLOW or BLOCK.
duration	TimeDuration	1	Duration of the time between updates of the IPPF security threat database.

Table 68-IpPortProtocolFilter Attributes

17.21 MalwareDetectionRemoval

Provides for the identification of Malware in an Application Flow and removal of the Malware or blocking of the subset of the Application Flow containing the malware. [Reference MEF 88 Section 6.8 Malware Detection and Removal Parameter.](#)

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
mdR_BlockList	Md_R	1..*	Supported Md_R list.
mdR_AllowList	Md_R	1..*	Unsupported Md_R list.
mdR_QuarantineList	Md_R	1..*	Block Md_R list.
allowList	Mbf	1..*	Allow MBF list.
nm	String String Enum: - ALLOW - BLOCK	1	Determines behavior when a subset of the Application Flow does not match a match criteria entry on any of the MD+R Lists.
duration	TimeDuration	1	Duration of time between updates of MD+R security thread database.
detection	String	1	Detection type.
removalBehavior	String String Enum: BLOCK_OBJECT_BLOCK_SUBSET_AF	1	Malware removal behavior.

	- BLOCK_OBJECT_ALLOW_BSET_AF - QUARANTINE_OBJECT_BLOCK_SUBSET_AF - QUARANTINE_OBJECT_ALLOW_SUBSET_AF - REMOVAL_MALWARE_ALLOWED_SUBSET_AF		
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Table 69-MalwareDetectionRemoval Attributes

17.22 Mbf

Maintained by the Service Provider for each Application Flow, which each match criteria entry on a list using a 2-tuple of the form <P, CS>, where P is a specific TLS protocol version and CS is the list of cipher suites for that TLS protocol version. Any can be used to indicate that any cipher suite for a given TLS protocol version is on the list.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
tlsProtocolVersion	String	1	TLS Protocol version.
cipherList	CipherSuite	1	Cipher suite.

Table 70-MbfList Attributes

17.23 MbfOrNone

Data type to allow the selection of MBF or None. Note that the value of None indicates that the MBF Security Function is Disabled for all Application Flows to which this Security Policy is applied.

Schema File Name: sdWan/sdWanCommon.yaml one of: -required: [mbfNone] -required: [mbfSelect]			
Attribute Name	Type	Multiplicity	Description
mbfNone	String Enum: NONE	1	Select NONE for no MBF.
mbfSelect	MiddleBoxFunction	1	Select MiddleBoxFunction.

Table 71-MbfOrNone Attributes

17.24 Md_R

Under development.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
	String	1	

Table 72-MdR Attributes

17.25 MdR_OrNone

Data type to allow the selection of Md_R or *None*. Note that the value *None* indicates that the MD+R Security Function is *Disabled* for all Application Flows to which this Security Policy is applied. Reference MEF 88 Section 6.8 Malware Detection and Removal Parameter.

Schema File Name: /sdWan/sdCommon.yaml one of: -required: [mdR_None] -required: [mdR_Select]			
Attribute Name	Type	Multiplicity	Description
mdRNone	String Enum: NONE	1	Select NONE for no MD+R.

mdRSelect	MalwareDetectionRemoval	1	Select MD+R.
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Table 73-MdR_OrNone Attributes

17.26 Middlebox Security Function Policy

The Middlebox Security Function Policy is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is decrypted/re-encrypted, Blocked, or passed through unchanged by the Middlebox Security Function. Reference MEF 138[17] Section 9.3 Middlebox Security Function Policy. NOTE: Additional parameters from

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
supportedList	Mbf	1..*	Supported MBF list.
unsupportedList	Mbf	1..*	Unsupported MBF list.
blockList	Mbf	1..*	Block MBF list.
allowList	Mbf	1..*	Allow MBF list.
nm	String String Enum: - ALLOW - BLOCK	1	Determines behavior when a subset of the Application Flow does not match a match criteria entry on either the MBF Block List or MBF Allow List.
certificateAuthority	String	1..*	The method for notifying the client in the Subscriber's network is beyond the scope of MEF 88 (at this timeO.
caBehaviorIfInvalid	String Enum: - ALLOW - BLOCK	1	Determines behavior of the MBF when the target server certificate is invalid.

Table 74-MiddleBoxFunction Attributes

17.27 OneWayMeanPacketDelayPrimary

Specifies the primary One-Way Mean Packet Delay parameters.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
threshold	TimeDuration	1	Threshold value. See MEF 70.2 Figure 16.
ceiling	TimeDuration	1	Ceiling value. See MEF 70.2 Figure 16.
remediation	TimeDuration	1	Remediation value. See MEF 70.2 Figure 16.

Table 75-OneWayMeanPacketDelayPrimary Attributes

17.28 OneWayMeanPacketDelayVariationPrimary

Specifies the primary One-Way Mean Packet Delay Variation parameters.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
threshold	TimeDuration	1	Threshold value. See MEF 70.2 Figure 16.
ceiling	TimeDuration	1	Ceiling value. See MEF 70.2 Figure 16.
remediation	TimeDuration	1	Remediation value. See MEF 70.2 Figure 16.

Table 76-OneWayMeanPacketDelayVariationPrimary Attributes

17.29 OneWayMeanPacketLossRatioPrimary

Specifies the primary One-Way Mean Packet Delay Variation parameters.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
threshold	Percentage	1	Threshold value. See MEF 70.2 Figure 16.
ceiling	Percentage	1	Ceiling value. See MEF 70.2 Figure 16.

remediation	Percentage	1	Remediation value. See MEF 70.2 Figure 16.
-------------	------------	---	--

Table 77-OneWayMeanPacketDelayVariationPrimary Attributes

17.30 OneWayMeanPacketDelaySecondary

Specifies the secondary One-Way Mean Packet Delay parameters.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
threshold	TimeDuration	1	Threshold value. See MEF 70.2 Figure 16.

Table 78-OneWayMeanPacketDelaySecondary Attributes

17.31 OneWayMeanPacketDelayVariationSecondary

Specifies the secondary One-Way Mean Packet Delay Variation parameters.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
threshold	TimeDuration	1	Threshold value. See MEF 70.2 Figure 16.

Table 79-OneWayMeanPacketDelayVariationSecondary Attributes

17.32 OneWayMeanPacketLossRatioSecondary

Specifies the secondary One-Way Mean Packet Delay Variation parameters.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
threshold	Percentage	1	Threshold value. See MEF 70.2 Figure 16.

Table 80-OneWayMeanPacketDelayVariationSecondary Attributes

17.33 PerformanceMetricsPrimary

Describes the most important Performance Metrics for this Application Flow for Primary.

Schema File Name: /sdWan/sdWanCommon.yaml			
---	--	--	--

Attribute Name	Type	Multiplicity	Description
oneWayMeanPacketDelay	OneWayMeanPacketDelayPrimary	1	Primary One-Way Mean Packet Delay definition.
oneWayMeanPacketDelayVariation	OneWayMeanPacketDelayVariationPrimary	1	Primary One-Way Mean Packet Delay Variation definition.
oneWayMeanPacketLossRatio	OneWayMeanPacketLossRatioPrimary	1	Primary One-Way Mean Packet Loss Ratio definition.

Table 81-PerformanceMetricsPrimary Attributes

17.34 PerformanceMetricsSecondary

Describes the most important Performance Metrics for this Application Flow for Secondary.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
oneWayMeanPacketDelay	OneWayMeanPacketDelaySecondary	1	Secondary One Way Mean Packet Delay definition.
oneWayMeanPacketDelayVariation	OneWayMeanPacketDelayVariationSecondary	1	Secondary One Way Mean Packet Delay Variation definition.
oneWayMeanPacketLossRatio	OneWayMeanPacketLossRatioSecondary	1	Secondary One Way Mean Packet Loss Ratio definition.

Table 82-PerformanceMetricsSecondary Attributes

17.35 PerformancePolicyCriterion

Allows the Subscriber to indicate the importance Performance Metrics for each Ingress Application Flow.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
primary	PerformanceMetricsPrimary	1	Primary performance metrics.
secondary	PerformanceMetricsSecondary	1	Secondary performance metrics.

Table 83-PerformancePolicyCriterion Attributes

17.36 Policy

Associated with each SWVC is a list of Policies that be assigned to each Application Flow. A Policy is composed of Policy Criteria. Reference MEF 70.2 Section 9.12 SWVC List of Policies Service Attribute.

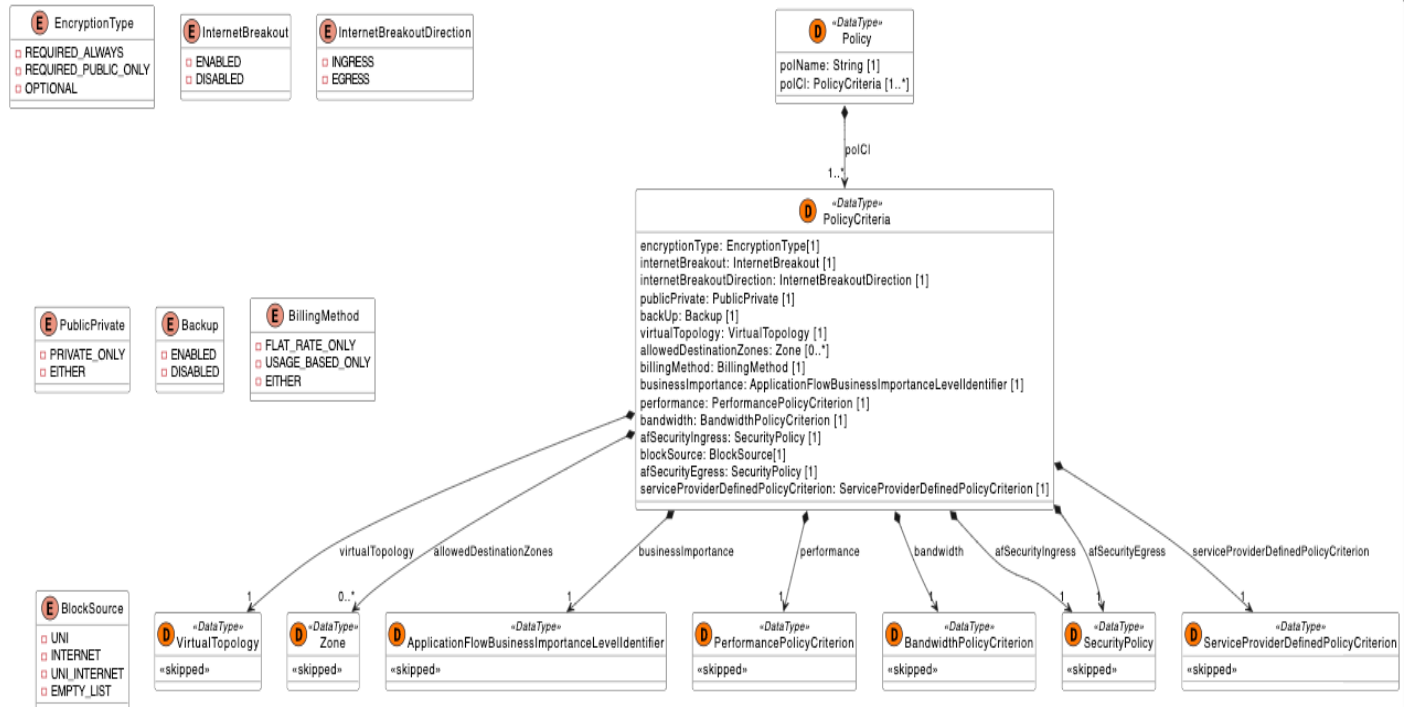


Figure 24-Policy Model

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
polName	String Enum: Not: - NONE - BLOCK	1	An identifier String that specifies the Policy.
polCl	PolicyCriteria	1..*	A non-empty list of Policy Criteria 2-tuples of form <pcName, pcParam>

Table 84-Policy Attributes

17.37 PolicyCriteria

Data type representing a Policy Criteria 2-tuples, of the form *<pcName,pcParam>*. Reference MEF 70.2 Section 9.12 SWVC List of Policies Service Attribute, Tables 4 and 5.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
encryptionType	String Enum: - REQUIRED_ALWAYS - REQUIRED_PUBLIC_ONLY - OPTIONAL	0..1	Indicates whether the Application Flow requires encryption.
internetBreakout	String Enum: - ENABLED - DISABLED	0..1	Indicates whether the Application Flow can be forwarded to an Internet destination.
publicPrivate	String Enum: - PRIVATE_ONLY - EITHER	0..1	Indicates whether the Application Flow can traverse Internet Access Underlay Connectivity Service or not.
backup	String Enum: - ENABLED - DISABLED	0..1	Indicates whether this Application Flow can use a UCS designated as “backup”.
virtualTopology	VirtualTopology	0..1	Indicates the Virtual Topology that the Application Flow should be forwarded over.

allowedDestinationZones	Zone	0..*	Specifies which Zones the IP Packets in the Application Flow can be delivered to.
billingMethod	String Enum: • FLAT_RATE_ONLY • USAGE_BASED_ONLY • EITHER	0..1	Indicates whether the Application Flow can be sent over an Underlay Connectivity Service that has usage-based or flat-rate billing.
businessImportance	ApplicationFlowBusinessImportanceLevelIdentifier	0..1	Indicates the relative business importance of the Application Flow.
performance	PerformancePolicyCriterion	0..1	Specifies a list of performance requirements for the Application Flow.
bandwidth	BandwidthPolicyCriterion	0..1	Specifies a bandwidth commitment and bandwidth limit on the Application Flow.
afSecurityIngress	SecurityFunctionPolicy	0..1	Specifies the Security Functions to apply to the Application Flow at Ingress UNI.
blockSource	String Enum: • UNI • INTERNET • UNI_INTERNET • EMPTY_LIST	0..1	Indicates specific source to disallow for egress. • UNI – Discard all IP Packets in the Application Flow if the Application Flow is not in Zone INTERNET. • INTERNET – Discard all IP Packets in the Application Flow if the Application Flow is in Zone INTERNET. • UNI_INTERNET – Discard all IP Packets in the Application Flow.

			EMPTY_LIST – No effect on IP Packets in the Application Flow.
afSecurityEgress	SecurityFunctionPolicy	0..1	Specifies the Security Functions to apply to the Application Flow at the Egress UNI.
serviceProviderDefinedPolicyCriterion	ServiceProviderDefinedPolicyCriterion	0..*	Service Provider defined own Policy Criteria.

Table 85-PolicyCriteria Attributes

17.38 PolicyMap

Specifies the Policies that are assigned to Application Flows and Application Flow Groups at the SWVC End Point. Reference MEF 70 Section 9.3 SWVC End Point Policy Map.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
applicationFlow	ApplicationFlowSpecification	0..1	Pointer to Application Flow. Reference MEF 70 Section 9.3 SWVC End Point UNI Service Attribute. NOTE: One of applicationFlow and applicationFlowGroup must be specified.
applicationFlowSpecificationGroup	ApplicationFlowSpecificationGroup	0..1	Pointer to Application Flow Group.
policy	Policy	1	Pointer to Policy.

Table 86-PolicyMap Attributes

17.39 SecurityPolicyIdentifier

Schema File Name: /sdWan/sdWanCommon.yaml

Parameter of a Security Policy that provides a unique identifier. Reference MEF 88 Section 6.1 Security Policy Identifier Parameter.

Type: string

maxLength: 53

pattern: "[/x20-\x7F]+"

17.40 ServiceProviderDefinedPolicyCriterion

Service Provider defined Policy Criterion. Reference MEF 70.2 Section 9.12 SWVC List of Policies Service Attribute.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
polName	string	1	Policy name.
criterionValue	string	1	Criterion value.
behavior	string	1	Behavior.
policyDirection	Enum: INGRESS EGRESS	1	Direction of the policy.

Table 87-ServiceProviderDefinedPolicyCriterion Attributes

17.41 UcsBillingMethod

File: /sdWan/sdWanCommon.yaml

UcsBillingMethod is an enumeration that indicates how access to the Underlay Connectivity Service is billed. Reference MEF 70.2 Section 12.1.3 UCS Billing Method Service Attribute.

- FLAT_RATE
- USAGE_BASE
- OTHER

17.42 UcsIdentifier

File: /sdWan/sdWanCommon.yaml

UCS Identifier as described in MEF 70.2 Section 12.1 UCS Identifier Service Attribute. A string that is used to allow the Subscriber and Service Provider to uniquely identify an Underlay Connectivity Service.

- type: string
- maxLength: 53
- pattern: "[/x20-\x7F]+"

17.43 UcsType

UcsType is an enumeration that represents the type of Underlay Connectivity Service types. Reference MEF 70.2 Section 12.1.2 UCS Type Service Attribute.

- PUBLIC
- PRIVATE

17.44 UnnamedRateLimiter

A 2-tuple, <name, commit, limit> where:

- commit is the threshold rate (bits per second) at or below which the SD-WAN Service Provider commits to deliver packets in the Bandwidth Flow with high probability under all traffic conditions.
- limit is the threshold information rate (bits per second) above which the Service Provider does not deliver IP Packet in the Bandwidth Flow under any conditions.

Reference MEF 70.2 Section 9.9 SWVC List of Rate Limiters Service Attribute.

File: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
commit	InformationRate	1	Threshold rate at or below which the SD-WAN Service Provider commits to deliver packets in the Bandwidth Flow with high probability under all traffic conditions.
limit	InformationRate	1	Threshold information rate above which the Service Provider does not deliver IP Packet in the Bandwidth Flow under any conditions.

Table 88-UnnamedRateLimiter Attributes

17.45 VirtualTopology

Attribute representing differing forwarding topologies to different Application Flows. The value is a list of 3-tuples of the for <vtName,vtType,vtEP>. Reference MEF 70.2 Section 9.6 SWVC List of Virtual Topologies Service Attribute.

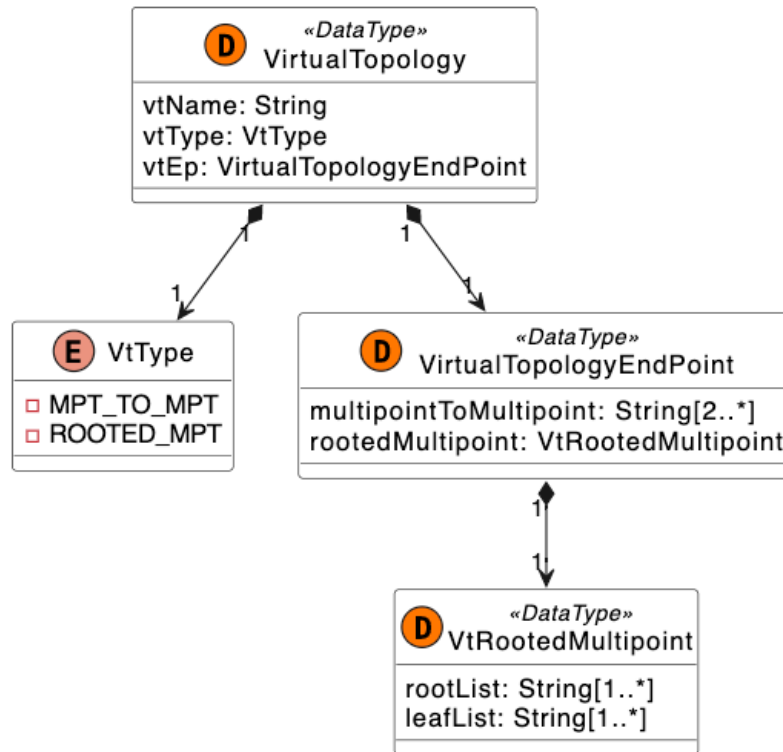


Figure 25-VirtualTopology Model

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
vtName	String	1	An identifier String that specifies the name of the Virtual Topology.
vtType	Enum: - MPT_TO_MPT - ROOTED_MPT	1	Provides a description of the type of connectivity provided by the Virtual Topology.
vtEp	VirtualTopologyEnd Point	1	Provides details of the SWVC End Points that are connected by the Virtual Topology.

Table 89-VirtualTopology Attributes

17.46 VirtualTopologyEndPoint

Virtual Topology End Point data type.

Schema File Name: sdWan/sdWanCommon.yaml

one of: -required: [multipointToMultipoint] -required: [rootedMultipoint]			
Attribute Name	Type	Multiplicity	Description
multipointToMultipoint	SwVcEndPointIdentifier	2..*	Multipoint-to-Multipoint for a Virtual Topology requires specifying at least two SWVC End Point Identifiers,
rootedMultipoint	VtRootedMultipoint	1	Rooted-Multipoint Virtual Topology.

Table 90-VirtualTopologyEndPoint Attributes

17.47 VtRootedMultipoint

Rooted-multipoint data type for a Virtual Topology.

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
rootList	SwVcEndPointIdentifier	1..*	Root list.
leafList	SwVcEndPointIdentifier	1..*	Leaf list.

Table 91-VirtualTopologyEndPoint Attributes

17.48 Zone

Provides a mechanism to identify the Zones supported by SD-WAN Service and to enumerate a set of Zone-side Policies for each Zone. Reference MEF 70.2 Section 9.5 SWVC List of Zones Service Attribute.

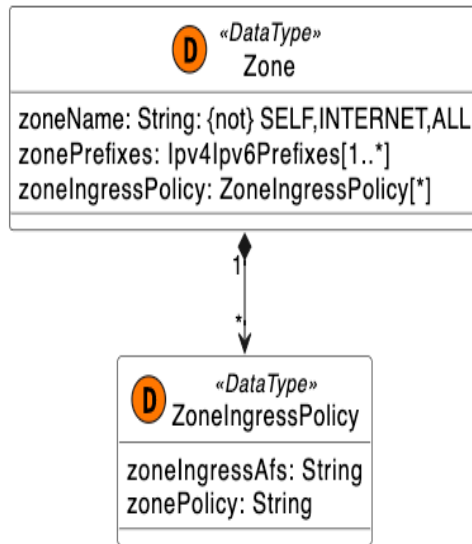


Figure 26-Zone Model

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
zoneName	String NOT(SELF, INTERNET, ALL)	1	An identifier String that specifies the name of the Zone.
zonePrefixes	Ipv4Ipv6Prefixes	1..*	Non-empty list of IP prefixes that identify the hosts in the Zone.
zoneIngressPolicy	ZoneIngressPolicy	0..*	A list (possibly empty) of 2-tuples <zoneIngressEFS, zoneIngressPol>.

Table 92-Zone Attributes

Schema File Name: sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
zoneIngressAfs	String	1	Application Flow Specification Group Name.
zonePolicy	String	1	Name of ingress policy.

Table 93-ZoneIngressPolicy Attributes

17.49 Urlf

Data type representing URL.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
url	String Format: uri	1	URL name.

Table 94-Urlf Attributes

17.50 UrlFilter

Provides for the filtering of Application Flows with respect to URLs. Reference MEF 88 Section 6.7 URL Filtering Parameter.

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
urlfBlockList	lppf	1..*	List of match criteria in URLF Block List or None.
urlfAllowList	lppf	1..*	List of match criteria in URLF Allow List or None.
urlfQuarantineList	lppf	1..*	List of match criteria in URLF Quarantine List or None.
nm	String	1	Parameter that determines the behavior when a subset of the Application Flow does not match criteria entry on any of the URLF List. The possible values are ALLOW or BLOCK.
duration	TimeDuration	1	Duration of the time between updates of the IPPF security threat database.

Table 95-UrlFilter Attributes

17.51 UrlfOrNone

Data type to allow the selection of URL Filter or None. Note that the value None indicates that the DNF Security Function is Disabled for all Application Flows to which this Security Policy is applied. Reference MEF 88 Section 6.7 URL Filtering Parameter.

Schema File Name: sdWan/sdWanCommon.yaml
one of:
-required: [urlfNone]

-required: [urlfSelect]			
Attribute Name	Type	Multiplicity	Description
urlfNone	String Enum: • NONE	1	Select NONE for no URL Filtering.
urlfSelect	UrlFilter	1	Select URL Filtering.

Table 96-UrlfOrNone Attributes

1529

18 Security Functions for IP Resources and Types

This section focuses on Security Functions for IP resources and types and defined in MEF 138 [17]. Data types and enumerations that are used by the SD-WAN Service model are defined below. The set of defined resources in support of Security Functions include:

- Security Action Lists: when a Security Function is enabled, a specific Block List, Allow List, and Quarantine List for that Security Function is required. The generic requirements related to the list is specified.
- Security Event Notification (SEN): notifies the agreed upon list of Subscriber personnel when security actions are taken.
- Security Admin Notification (SAN): notifies the agreed upon list of Subscriber personnel when a significant change to a Security Function Policy is made.
- DNS Resolution: provides the capability to offer a service responding to customer DNS request with either upstream DNS responses (from the legitimate DNS servers for a given resource) or to examine the request and insert a different customer response based on the security policy applied by the various security functions.

Additional Security Functions as defined in MEF 138[17] are listed:

• Middlebox Security Function

- IP, Port and Protocol Filtering, when enabled for a given Service Flow, can Allow or Block a subset of the Service Flow based on the source or destination IP address, source or destination port number, and/or IP protocol.
- Domain Name Filtering, when enabled for a given Service Flow, can Allow or Block a subset of the Service Flow based on domain name. Wildcard entries are supported.
- URL Filtering, when enabled for a given Service Flow, can Allow or Block a subset of the Service Flow based on the URL. Wildcard entries are supported.
- Malware Detection and Removal, when enabled for a given Service Flow, scans Objects for Malware and takes appropriate action when Malware is detected.

- Data Loss Prevention, when enabled can protect the Service Flow from possible exfiltration of Personally Identifiable Information (PII) and Confidential and Proprietary Information (CPI).
- DNS Protocol Filtering, when enabled for a given Service Flow, can Allow or Block a subset of the Service Flow based on DNS message type and source/destination IP addresses.
- Protective DNS, when enabled for a given Service Flow, provides for enhanced security for DNS, i.e., checking whether each DNS request/response is allowed.

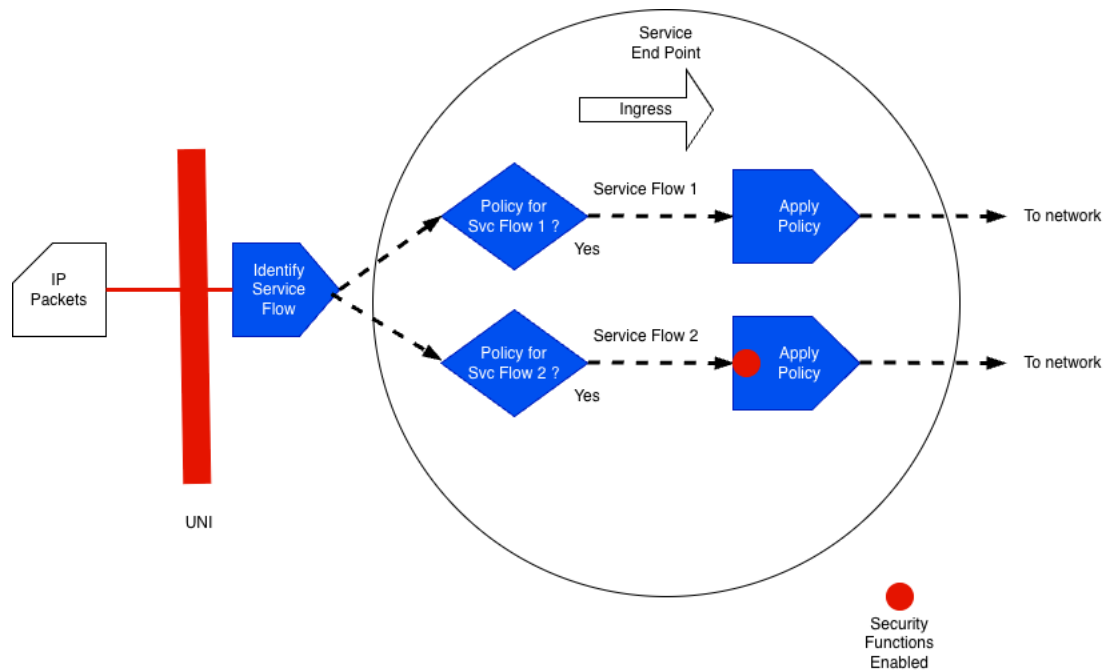


Figure 27-Service Flow Security: Ingress Service Flow Example

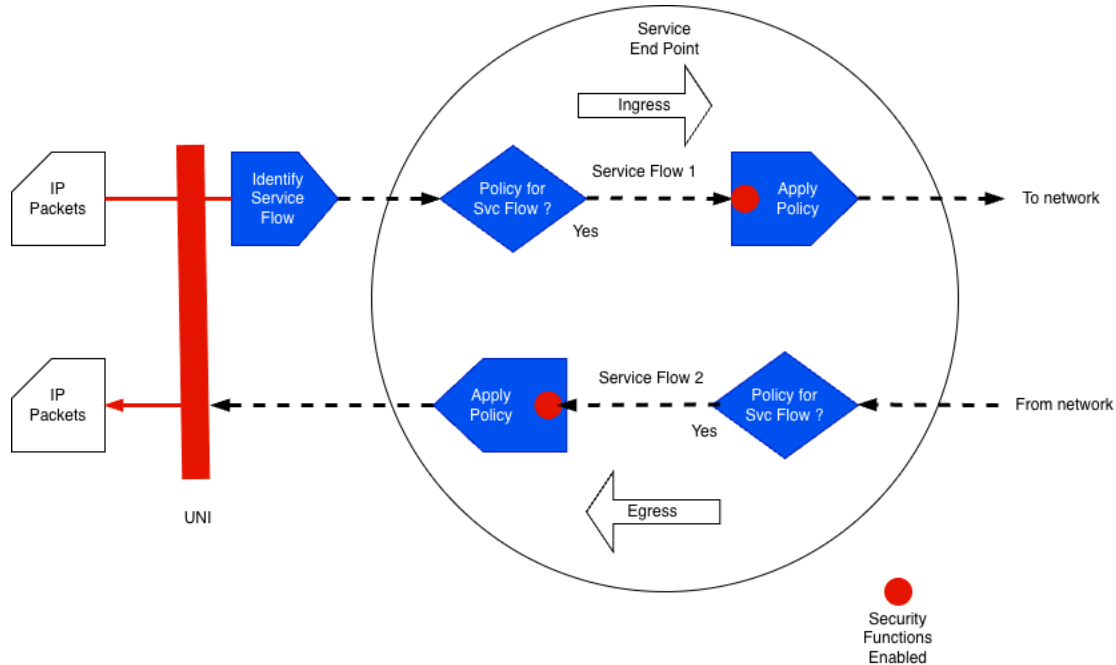


Figure 28-Relationship of Security Functions and Service Flows Example

9.10 SWVC List of Security Policies Service Attribute

The value of the SWVC List of Security Policies Service Attribute is a list (which may be empty) of Security Policies that are available for use with this SWVC. MEF 88 [38] section 6 defines the structure and content of each Security Policy in the SWVC List of Security Policies Service Attribute. The Security Policies are assigned to Ingress and Egress Application Flows via the AF-SECURITY-INGRESS Policy Criterion (section 9.12.2.11) and the AF-SECURITY-EGRESS Policy Criterion (section 9.12.3.2). One of the Security Policy parameters defined in MEF 88 [38] is the Security Policy Identifier.

18.1 SecurityFunctionPolicy

A set of parameters that are agreed between the Subscriber and Service Provider (as part of the SWVC List of Security Function Policies Service Attribute) and that specify which Security Function Policies are to be applied to an Application Flow. A Security Policy could be referenced in zero or more Ingress (SD-WAN) Policies and zero or more Egress (SD-WAN) Policies. Reference MEF 138 Section 9 Security Function Policy.

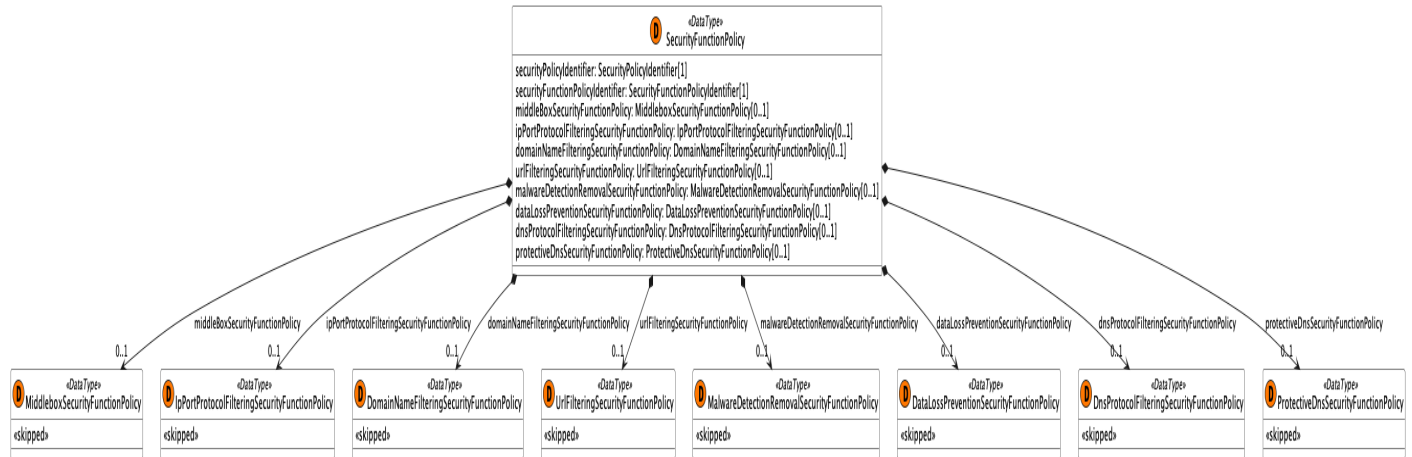


Figure 29-Security Function Policy Model

Schema File Name: /sdWan/sdWanCommon.yaml			
Attribute Name	Type	Multiplicity	Description
securityPolicyIdentifier	SecurityPolicyIdentifier	1	Parameter of a Security Policy that provides a unique identifier. Reference MEF 138 Section 9.1 Security Policy Identifier Parameter.
securityFunctionPolicyIdentifier	SecurityFunctionPolicyIdentifier	1	Notification to the Subscriber of a security event. Reference MEF 138 Section 9.2 Security Function Policy Identifier Parameter.
middleboxSecurityFunctionPolicy	MiddleboxSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is decrypted/re-encrypted, Blocked, or passed through unchanged by the Middlebox Security Function. Reference MEF 138, Section 9.3 Middlebox Security Function Policy.
ipPortProtocolFilteringSecurityFunctionPolicy	IpPortProtocolFilteringSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subset of Service Flow is Allowed, Blocked, or Quarantined by the IPPF Security Function. Reference MEF 138, Section 9.4 IP, Port and Protocol

			Filtering (IPPF) Security Function Policy.
domainNameFilteringSecurityFunctionPolicy	DomainNameFilteringSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is Allowed, Blocked, or Quarantined by the DNF Security Function. Reference MEF 138, Section 9.5 Domain Name Filtering (DNF) Security Function Policy.
urlFilteringSecurityFunctionPolicy	UrlFilteringSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is Allowed, Blocked, or Quarantined by the URLF Security Function. Reference MEF 138, Section 9.6 URL Filtering (URLF) Security Function Policy.
malwareDetectionRemovalSecurityFunctionPolicy	MalwareDetectionRemovalSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is Allowed, Blocked, or Quarantined by the MD+R Security Function. Reference MEF 138, Section 9.7 Malware Detection and Removal (MD+R) Security Function Policy.
dataLossPreventionSecurityFunctionPolicy	DataLossPreventionSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subnet of the Service Flow is Allowed, Blocked, or Quarantined by the DLP Security Function. Reference MEF 138 Section 9.8 Data Loss Prevention (DLP) Security Function Policy.
dnsProtocolFilteringSecurityFunctionPolicy	DnsProtocolFilteringSecurityFunctionPolicy	1	Is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is Allowed, Blocked, or Quarantined by the DPF Security Function. Reference MEF 138 Section 9.9 DNS Protocol

			Filtering (DPF) Security Function Policy.
protectiveDnsSecurityFunctionPolicy	ProtectiveDnsSecurityFunctionPolicy	0..1	Is an Atomic Policy that defines the rules that determine whether a subset of the Service Flow is Allowed, Blocked, or Quarantined by the PDNS Security Function. Reference MEF 138 Section 9.10 Protective DNS Security Function Policy.

Table 97-SecurityFunctionPolicy Attributes

18.2 Security Event Notification (SEN)

A Security Event is defined as an incident associated with a Service Flow detected by a Security Function that triggers a notification to the Subscriber due to either a) a Notification Action Modifier associated with the criteria entry on an of the Security Function's Lists, or b) the Service Flow does not match a criteria entry on any of the Security Function's Lists but is Blocked due to the application of the Security Function Policy.

Figure 30-Security Event Notification Model

18.3 SecurityEventNotification

Data type to allow the selection of SecurityEventNotification. A SEN is a notification to the Subscriber of a security event. The value of SEN is either *None* or a two-tuple of the form (R, T) where R is a list of recipients authorized to receive the SEN, and T is the format of the timestamp of the SEN, e.g., data and time in UTC or local time. **Reference MEF 88 Section 6.2 Security Event Notification Parameter.**

Schema File Name: /sdWan/sdWanCommon.yaml

Attribute Name	Type	Multiplicity	Description
receipients	Receipient	1	Receipients of security event notifications.
timestamp	TimeAndDate	1	Timestamp.

Table 98-SecurityEventNotification Attributes

18.4 SecurityEventNotificationOrNone

Data type to allow the selection of SecurityEventNotification or *None*. Note that the value *None* indicates that the SEN if not need for this Security Policy. Reference MEF 88 Section 6.2 Security Event Notification Parameter.

Schema File Name: /sdWan/sdCommon.yaml one of: -required: [ippfNone] -required: [ippfSelect]			
Attribute Name	Type	Multiplicity	Description
senNone	String Enum: • NONE	1	Select NONE for no SEN.
senSelect	SecurityEventNotific ation	1	Select SEN.

Table 99-SecurityEventNotificationOrNone Attributes

18.5 Security Admin Notification (SAN)

A Security Admin Notification (SAN) is a notification to the agreed upon list of Subscriber personnel of a change to a Security Function Policy.

Figure 31-Security Admin Notification Model

19 References

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- 1647 [13] MEF 70.2 *SD-WAN Service Attributes and Service Framework*, October 2023.
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- 1649 [15] MEF 105 *Performance Monitoring and Service Readiness Testing for SD-WAN Service*,
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1652 2023.
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Appendix A Usage examples (Informative)

This appendix aims to provide an extensive set of examples to cover:

- Different Service Order configuration variants,
- Basic Service Order API walkthrough to order a SD-WAN Service,
- Common modifications,
- Deletion of a Service
- List of Service Orders

The full set of examples are delivered in this document and as a Postman collection file available at:

- /documentation/serviceSchema/sdWan/Postman/Mplify_99_1_with_Mplify_100_Service_Ordering_Management.postman_collection.json

A.1 High-level Flow

The Interface Reference Points each form a set of APIs that service different functions in the end-to-end flow.

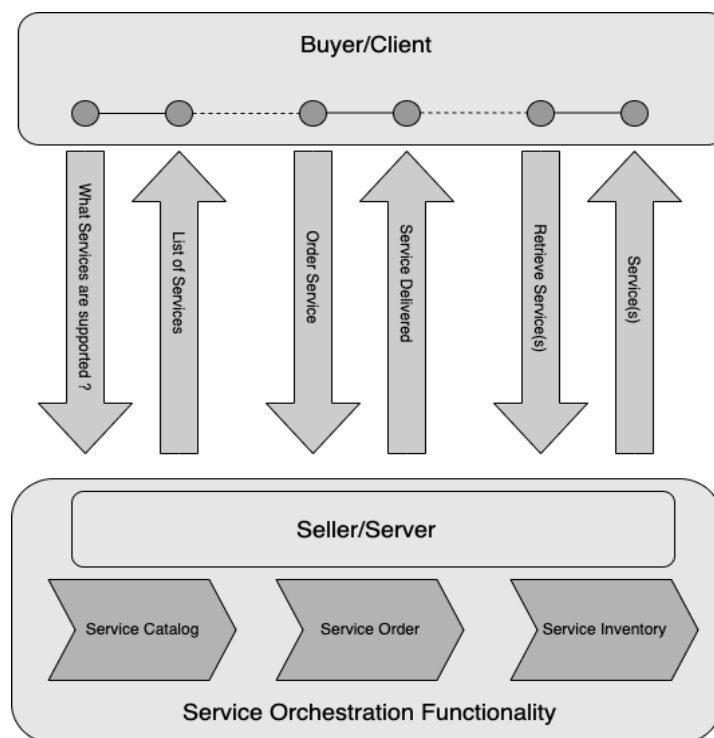


Figure 32-Service End-to-End Flow

Service Catalog – allows the Client to query SOF for available Services as well as what attributes are fixed and/or elastics with values/ranges.

Service Order – allows the Client to request the SOF to initiate and complete the fulfillment process of installation of a Service Offering, an update to an existing Service, or a disconnect of an existing Service.

Service Inventory – allows the Client to retrieve information about existing Service instances from the SOF's Service Inventory.

All the above-mentioned APIs are provided in the SDK together with accompanying Developer Guides. Please refer to those documents for more details and examples of functional APIs.

A.2 Integration of Service Specification into the Service Order API

The Service Order API is service-agnostic in the meaning that they serve as an interaction between the Client and the Server (SOF) and they do not contain any service-specific information in their specifications. To pass the service-specific information, an extension pattern is used. This applies to any of the Service APIs that carry service-specific information: Service Catalog, Service Order and Service Inventory.

The extension hosting type in the API data model is `MefServiceConfiguration`. The `@type` attribute of that type must be set of a value that uniquely identifies the service specification. See Figure 33. A unique identifier for Mplify standard service specifications is in URN format and is assigned by Mplify. This identifier is provided as root schema `$id` and in service specification documentation.

The example below shows a header of a SD-WAN UNI, where `$id: urn:mef:lso:spec:service:sd-wan-uni:v0.0.3:all` the above-mentioned URN:

```
$id: urn:mef:lso:spec:service:carrier-sd-wan-uni:v0.0.3:all
$schema: http://json-schema.org/draft-07/schema#
title: Mplify LSO Service SD-WAN UNI Service Specification
```

The extension pattern specific to SD-WAN Services is illustrated in Figure 33. The SD-WAN Service Extension model includes URN representations for `SdWanUni`, `Swvc`, `SwvcEndPoint`, `UcsUni`, `Ucs`, `UcsEndPoint`. The URNs are:

- `urn:mef:lso:spec:service:sd-wan-uni:v0.0.3:all`
- `urn:mef:lso:spec:service:swvc:v.0.0.3:all`

- 1700 • urn:mef:lso:spec:service:swvc-end-point:v.0.0.3:all
- 1701 • urn:mef:lso:spec:service:ucs:v.0.0.3:all
- 1702 • urn:mef:lso:spec:service:ucs-end-point:v.0.0.3:all
- 1703 • urn:mef:lso:spec:service:ucs-uni:v.0.0.3:all

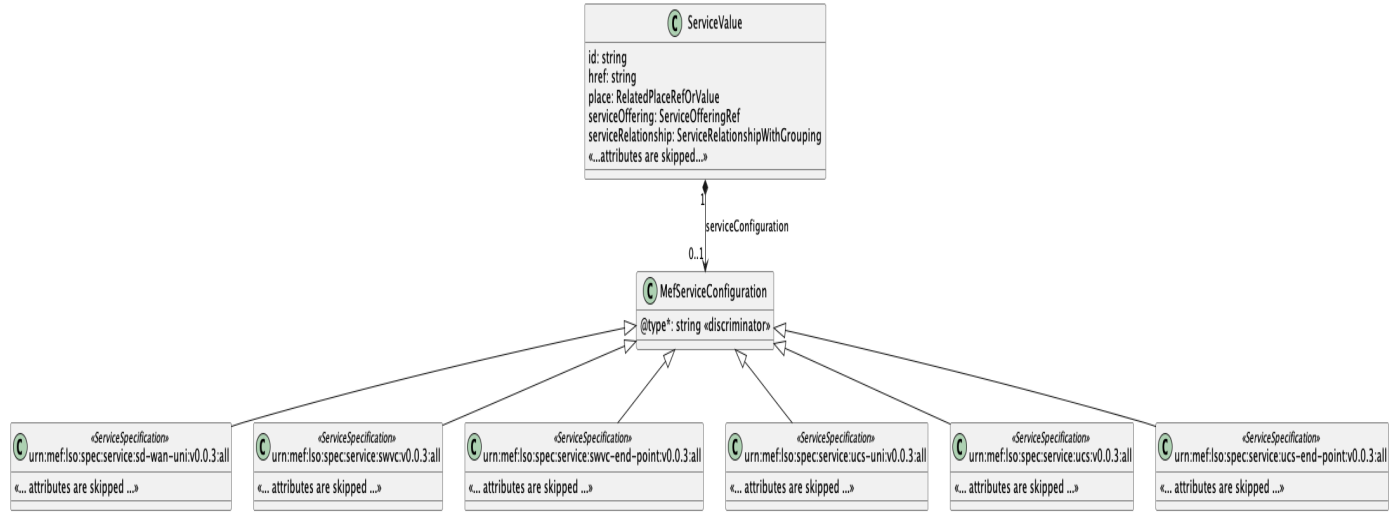


Figure 33-Extension Pattern: SD-WAN

Use of non-Mplify standard service definitions is allowed. In such a case the schema identifier must be agreed upon between the Client and the SOF.

A.3 Service Order Create Process

The **ServiceOrderCreate** process is initiated by the Buyer/Client with the determination of the one or more **ServiceOrderItems** that will be part of a Service Order using the **ServiceOrderCreate**. Each Service Order Item through the Extension patter is associated with a Mplify Service specific payload component (i.e., **SdWanUni**, **Swvc**, **SwvcEndPoint**, **UcsUni**, **Ucs**, **UcsEndPoint**).

The Buyer/Client sends a request with a **ServiceOrderCreate** type in the body. The SOF performs request validation, assigns and **id**, and returns **ServiceOrder** type in the response body, with a state set to **acknowledged**. From this point, the **ServiceOrder** is ready for further processing. The Buyer/Client can track the progress of the process by either subscribing for notifications (Asynchronous operation) or by periodically polling (Synchronous operation) the **ServiceOrder**. The two patterns are illustrated in sequence diagrams below.

1720 **A.3.1 Interaction Flow – Polling process (Synchronous)**

1721 The sequence flow for the Synchronous or Polling process begins with a Buyer/Client initiating a
1722 build request for a `ServiceOrder_Create` with one or more `ServiceOrderItems`.
1723 `ServiceOrderItems` include `SdWanUni`, `Swvc`, `SwvcEndPoint`, `UcsUni`, `Ucs`, `UcsEndPoint`
1724 and associated relationships between resources.

1725 The sequence begins with the Buyer/Client building the `ServiceOrder_Create` including the set
1726 of `ServiceOrderItems`. In the Synchronous operation of the API once the REST POST operation
1727 is sent the Buyer/Client needs to poll (REST GET) to determine the state of the request.



Figure 34-Service Order Polling process (Synchronous)

NOTE: A complete SD-WAN Service Order would also include one or more Underlay Connectivity Services. Each Underlay Connectivity Service would include `UcsUni(s)`, `UcsEndPoint(s)` and `Ucs(s)`.

1733 **A.3.2 Interaction Flow – Notification process (Asynchronous)**

1734 The sequence flow for the Asynchronous or Notification process begins with the Buyer/Client
1735 subscribing to Service Order asynchronous events. The Buyer/Client informs the Seller/Server the
1736 callback URL to send notifications.

1737 The Buyer/Client build a request for a `ServiceOrder_Create` with one or more
1738 `ServiceOrderItems`. `ServiceOrderItems` include `SdWanUni`, `SwvcEndPoint` and `Swvc` along
1739 with associated relationships between resources.

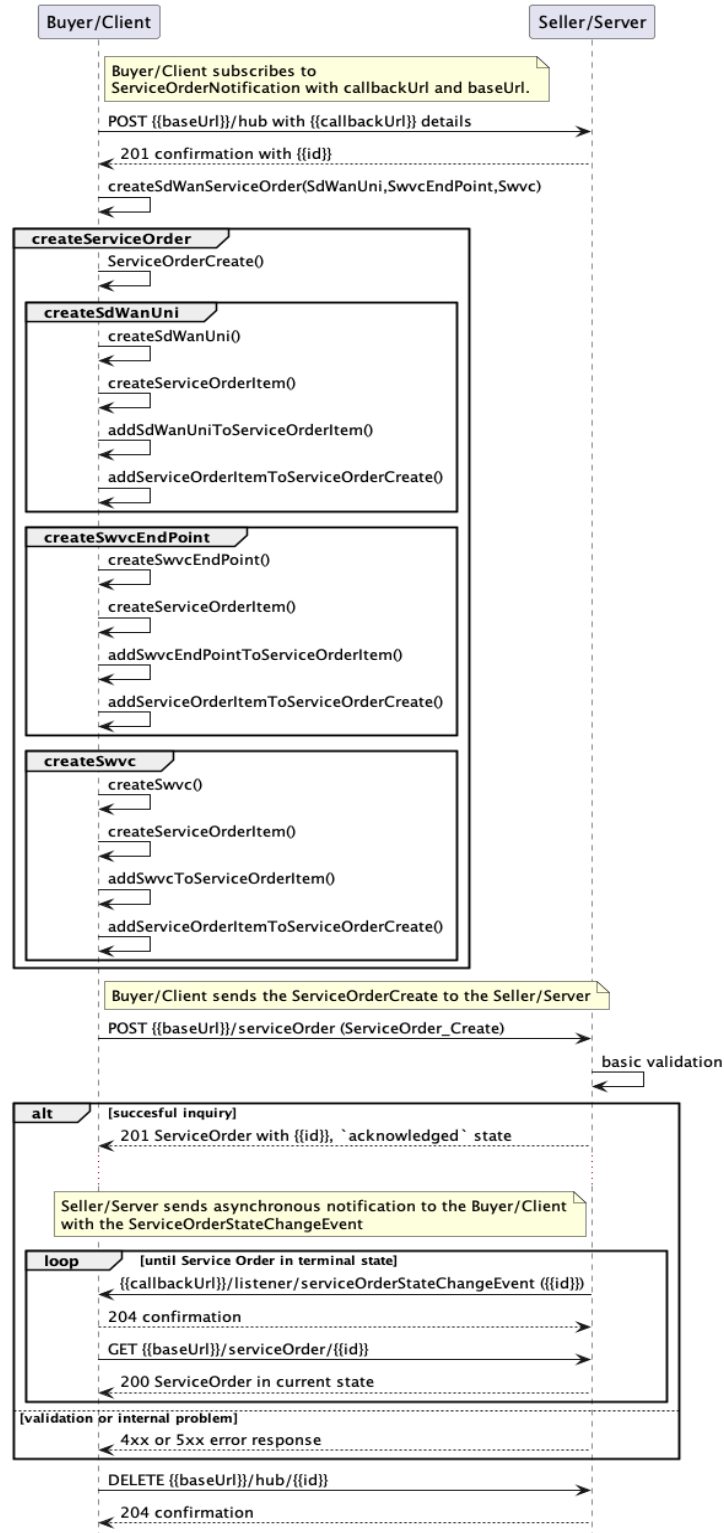


Figure 35-Service Order Notification process (Asynchronous)

NOTE: A complete SD-WAN Service Order would also include one or more Underlay Connectivity Services. Each Underlay Connectivity Service would include `UcsUni(s)`, `UcsEndPoint(s)` and `Ucs(s)`.

Appendix B Service Order Operations

The following section provides details on the set of Service Order operations that is support with the Mplify 99.1 Service Ordering Management API (Envelope) and Mplify W100 SD-WAN Service Schemas and Developer Guide (Payload). The current set of operations available include:

- GET /serviceOrder – List or find Service Order objects.
- POST /serviceOrder – Create a Service Order.
- GET /serviceOrder/{id} – Retrieves a Service Order by ID.
- POST /hub – Register a listener.
- DELETE /hub/{id} – Unregister a listener.
- GET /hub/{id} – Retrieve an Event Subscription by ID.

The remainder of this section will provide detailed JSON examples of the REST operations and responses.

B.1 UC1: GET /serviceOrder and Response

The following section provides an example for a retrieval of list of Service Orders given query parameters: state, orderDate, completionDate, expectedCompletionDate, startDate. The request is:

```
GET /{{{baseUrl}}}/mefApi/legato/serviceOrderingManagement/v6/serviceOrder
```

The following example below shows the response to a Service Order list request. The response as a `SdWanUni`, `SwvcEndPoint` and `Swvc`.

Editor Note 4: [PlantUML JSON]

Figure 36-List Service Order Response

The following snippet presents the actual JSON for the example Service Order List response.

Editor Note 5: [JSON]

B.2 UC2: POST /serviceOrder and Response

The following section provides an example of a Service Order Create with all support actions (ADD, MODIFY, DELETE) and corresponding response. The payload for the request is shown in Figure 37. The example is for a SD-WAN service. The REST request:

```
POST /{{baseUrl}}/mefApi/legato/serviceOrderingManagement/v6/serviceOrder
```

The Service Order Request show in Figure 37 is a partial request for the diagram to be readable. See the complete request in JSON following the JSON figure that follows.

Editor Note 6: [PlantUML JSON]

Figure 37-Service Order Request-action=ADD

The following snippet presents the actual JSON for the example Service Order Create request. There are three (3) Service Order Items: `SdWanUni`, `SwvcEndPoint` and `Swvc`. An actual Service Order could include one or more Underlay Connectivity Services and associated resources which are not shown.

Editor Note 7: [JSON]

NOTE: The Service Order request can contain relationships between Service Order Items (serviceOrderItemRelationship) within the Service Order as well as between a Service Order Item and an Inventory component (serviceRelationship).

The SOF response to the `CreateServiceOrder` is a `ServiceOrder` which is shown below. The main types used for the response are `ServiceOrder` and `ServiceOrderItem`. The one or more `ServiceOrderItems` will reference a specific SD-WAN resource as part of the payload and association(s) to other `ServiceOrderItems` as an attribute in the envelope. The response echoes back all the attributes as provided by the Client/Buyer and contains the same number of `ServiceOrderItems` as in the request.

The Service Order Request Response show in Figure 38 is a partial request for the diagram to be readable. See the complete request in the JSON following the JSON figure that follows.

Editor Note 8: [PlantUML JSON]

Figure 38-Service Order Response

The following snippet presents the actual JSON for the example Service Order Create response. There are five (3) Service Order Items for SD-WAN Service: `SdWanUni`, `SwvcEndPoint`, and `Swvc`. In an actual SD-WAN Service there would likely be one or more Underlay Connectivity Services.

B.2.1 UC2b: Action-Modify

This section guides through all the steps of Service Order API that is needed to be performed to successfully modify a service.

NOTE: SOF is free to mandate some of these steps.

NOTE: As the examples of steps in many cases will replicate the service-specific information, in some of the snippets some parts of it will be omitted for better readability.

There are rules for all request items for creation requests (Service Order):

- `item.action` must be set to `modify`
- `item.service.id` must be provided.
- `service.serviceConfiguration` must contain all desired configurations.

B.2.2 UC2c: Action-Delete

This section guides through all the steps of Service Order API that is needed to be performed to successfully delete a service. The action “delete” only moves the Service to the ‘terminated’ state and does not directly remove the Service from the SOF’s Service Inventory.

NOTE: SOF is free to mandate some of these steps.

There are rules for all request items for creation requests (Service Order):

- `item.action` for each Service Order Item must be set to `delete`

B.3 UC3: GET /serviceOrder/{id} and Response

The following section provides an example for a retrieval of a Service Order with a unique identifier. The REST request is:

```
1820 GET /{{baseUrl}}/mefApi/legato/serviceOrderingManagement/v6/serviceOrder/{id}
```

1821

1822 *Editor Note 9: [PlantUML JSON]*

1823 Figure 39- GET Service Order by ID Response

1824 The following snippet presents the actual JSON for the example Service Order by ID response.
1825 There is one `SdWanUni` in the response.

1826 *Editor Note 10: [JSON]*

1827 B.4 UC4: POST /hub and Response

1828 The following section provides an example register for a Service Order Management notification
1829 service from the Seller/Server. The REST POST request with REST Body is:

```
1830 POST /{{baseUrl}}/mefApi/legato/serviceOrderingManagement/v6/hub/
1831 {
1832   "callback": "clientUri://",
1833     "query":
1834     "eventType=
1835       serviceOrderCreateEvent,
1836       serviceOrderStateChangeEvent,
1837       serviceOrderInformationRequiredEvent,
1838       serviceOrderItemStateChangeEvent"
1839 }
```

1840 The REST successful (201 Created) Response is shown below:

```
1841 {
1842   "callback": "clientUri://",
1843     "id": "Event-Id-10",
1844     "query":
1845     "eventType=
1846       serviceOrderCreateEvent,
1847       serviceOrderStateChangeEvent,
1848       serviceOrderInformationRequiredEvent,
1849       serviceOrderItemStateChangeEvent"
1850 }
```

B.5 UC5: DELETE /hub/{id} and Response

The following section provides an example unregister for a Service Order Management notification service from the Seller/Server. The REST DELETE request is:

```
DELETE /{{baseUrl}}/mefApi/legato/serviceOrderingManagement/v6/hub/{id}
```

A successful DELETE will have a REST Response of 204 (No Content).

B.6 UC6: GET /hub/{id} and Response

The following section provides an example retrieval for a Service Order Management Event Subscription by unique identifier. The REST GET request is:

```
GET /{{baseUrl}}/mefApi/legato/serviceOrderingManagement/v6/hub/{id}
```

The REST successful (200 Success) Response is shown below:

```
{
  "callback": "clientUri://",
    "id": "100",
    "query":
    "eventType=
      serviceOrderCreateEvent,
      serviceOrderStateChangeEvent,
      serviceOrderStateChangeEvent,
      serviceOrderInformationRequiredEvent"
}
```

Appendix C Acknowledgements

The following contributors participated in the development of this document and have requested to be included in this list.

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